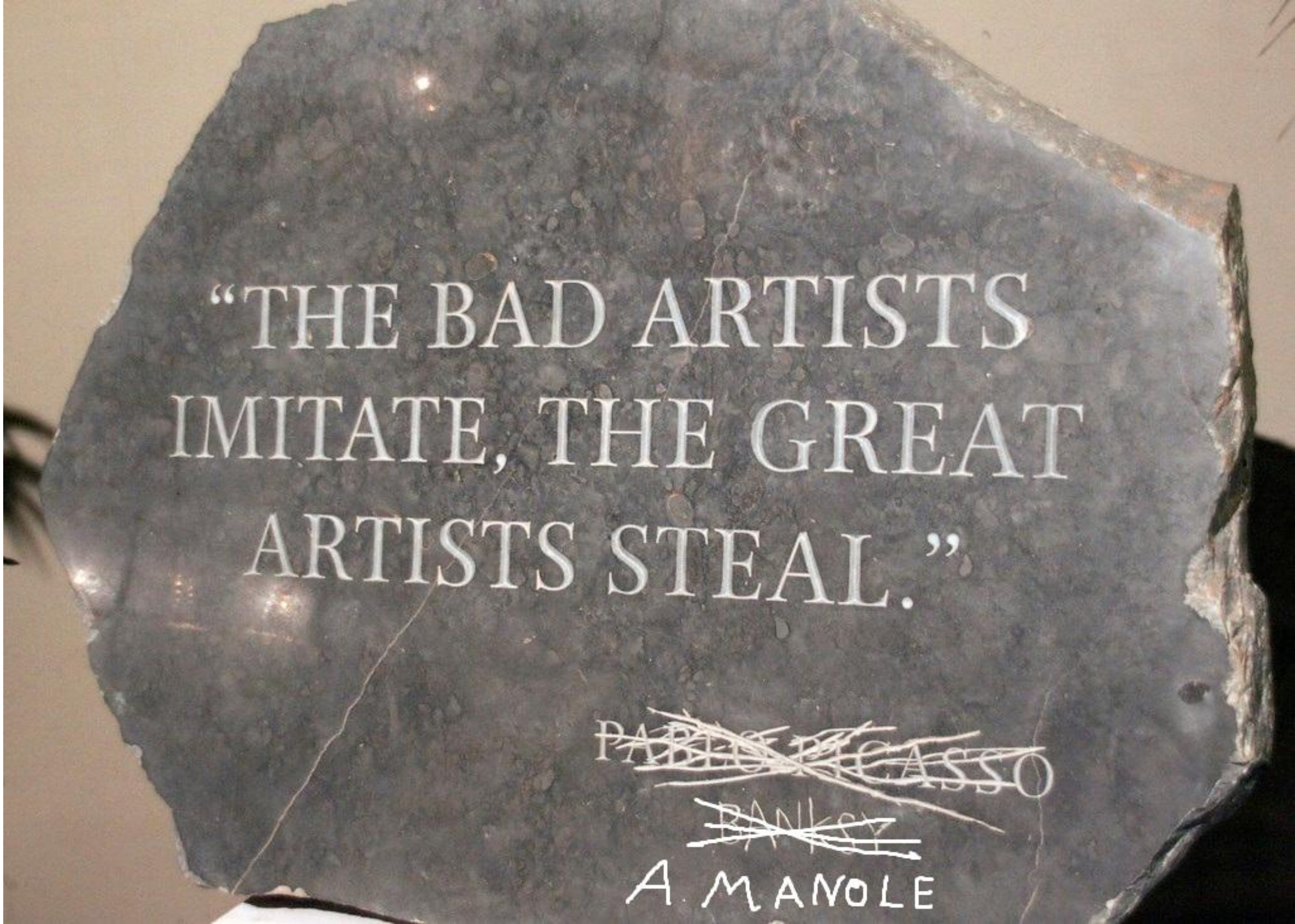
Stable Diffusion

ITSG - 2025

Introduction - Rabbit Hole

Rabbit Hole
Progress Tracking



Rabbit Hole
Progress Tracking

Instructions

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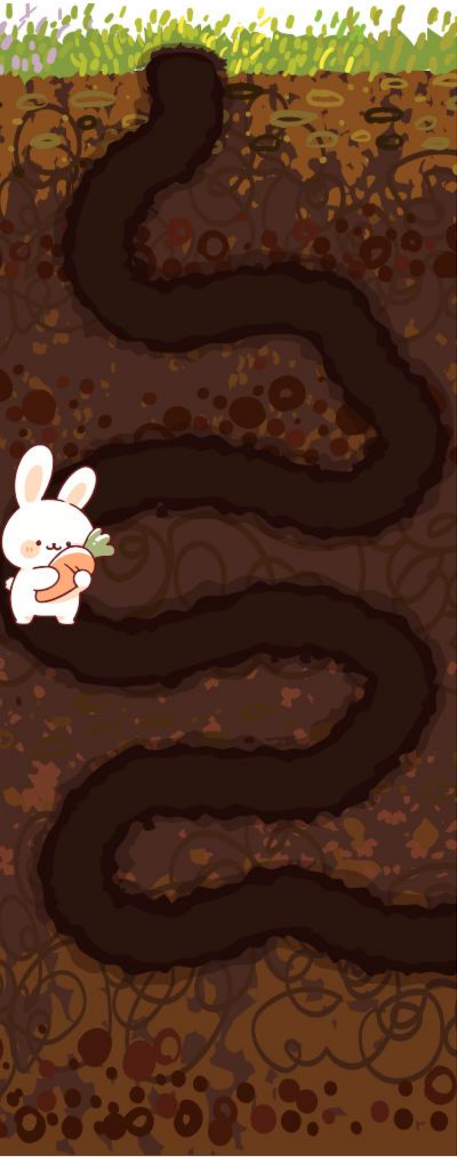
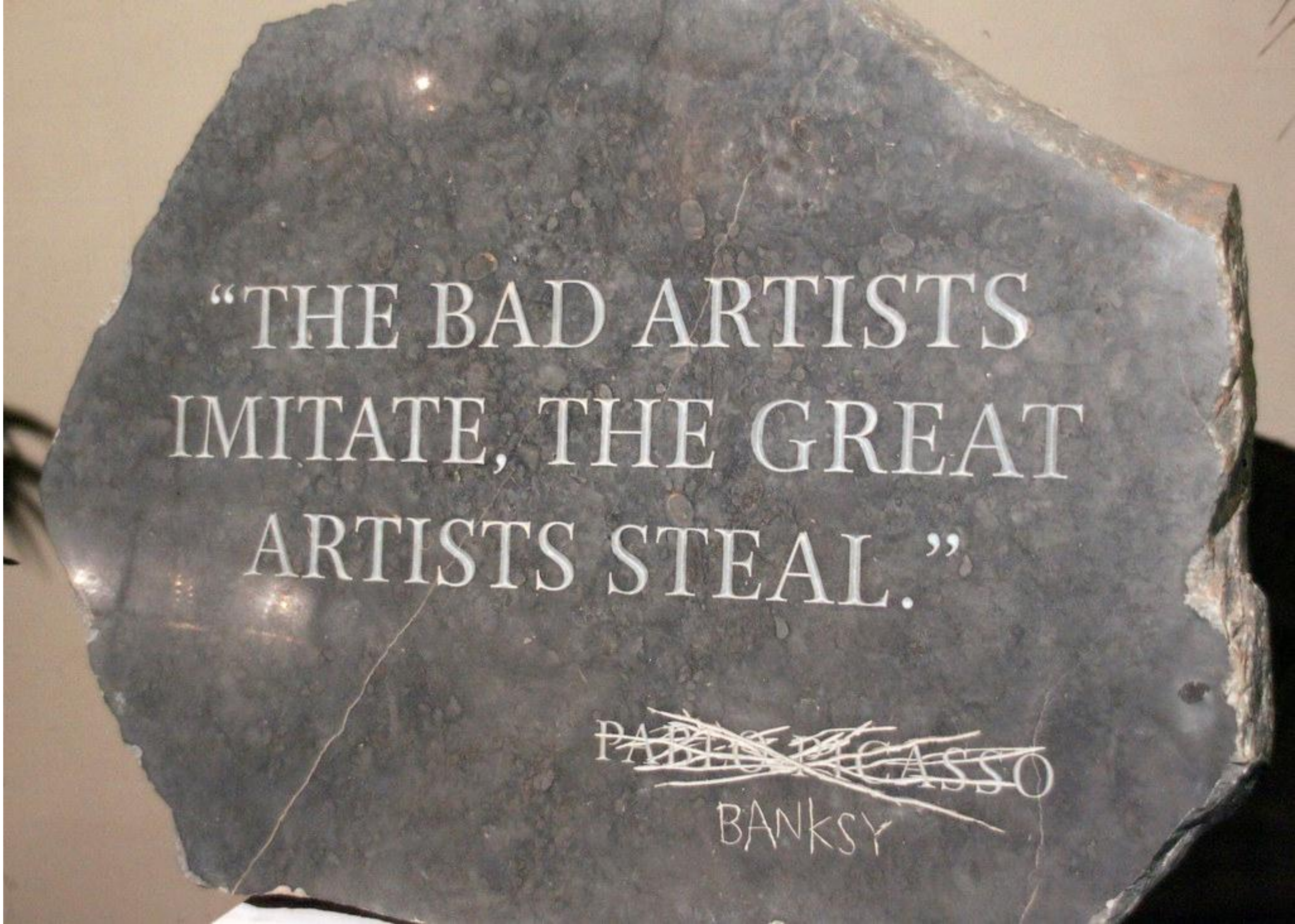
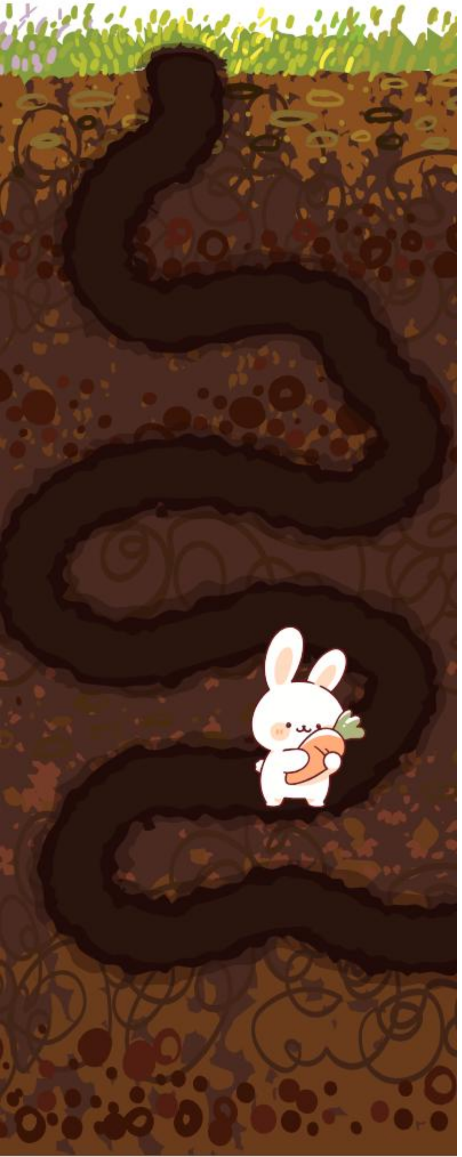
Or use QR code

Rabbit Hole
Progress Tracking

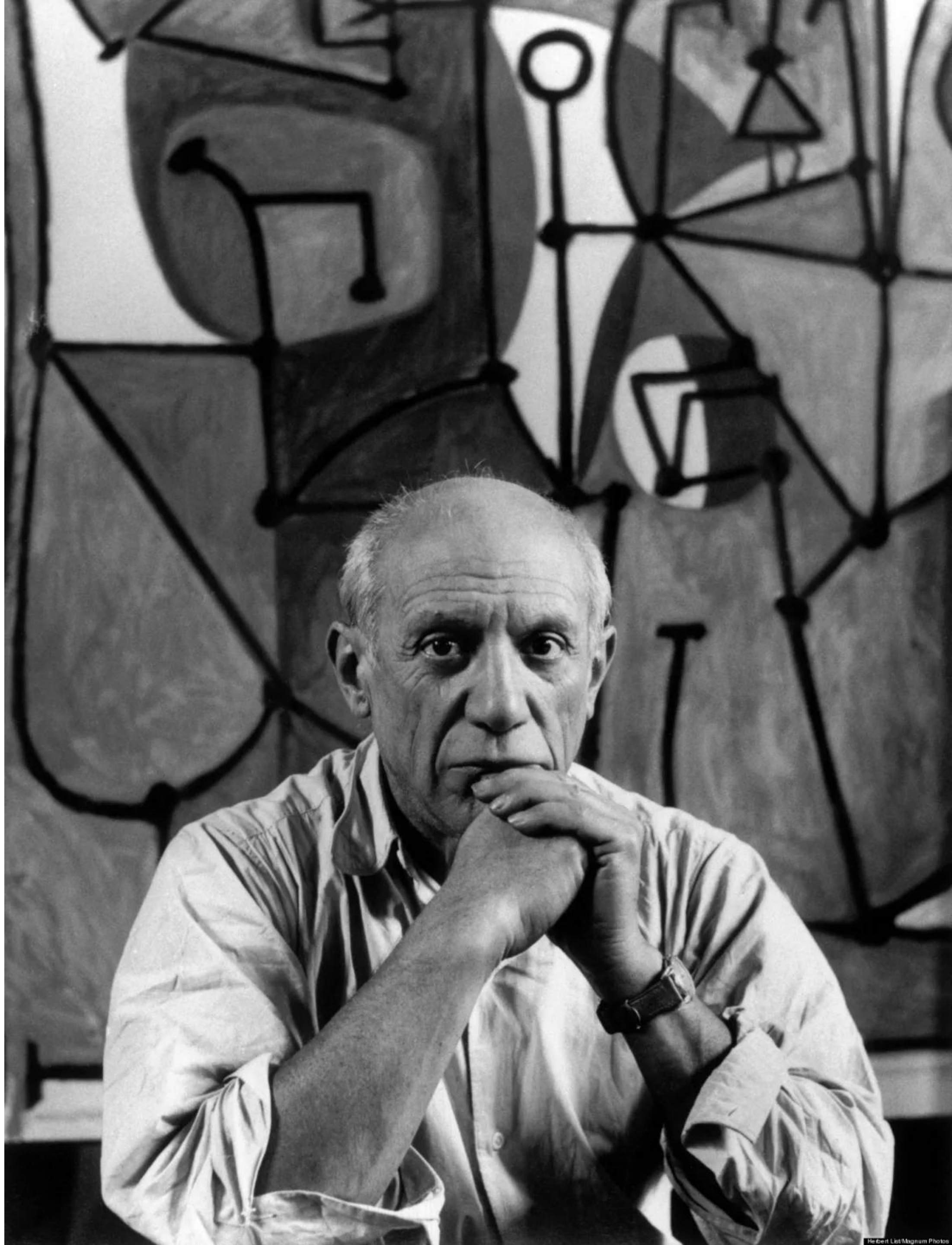




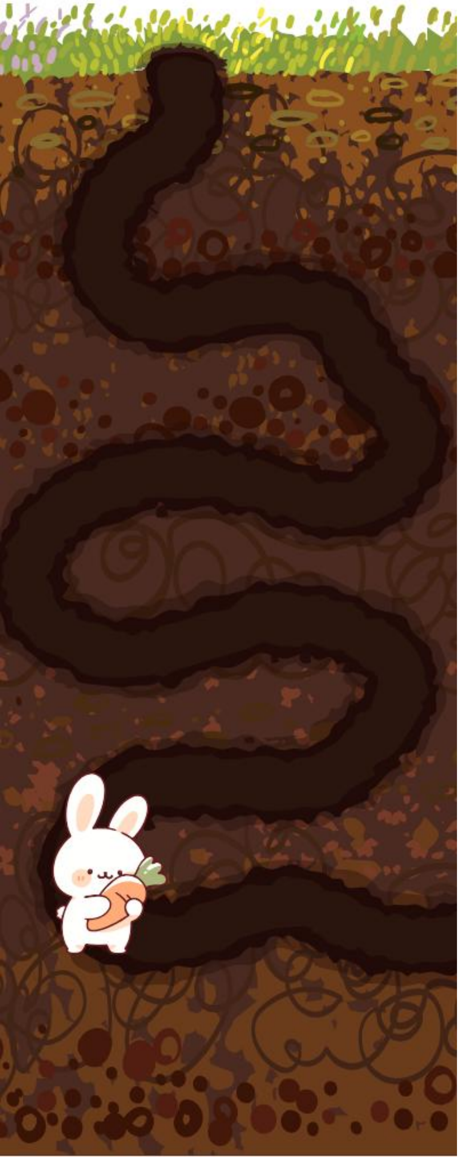
Rabbit Hole
Progress Tracking



"Good artists copy,
great artists steal."
- Pablo Picasso



Rabbit Hole
Progress Tracking

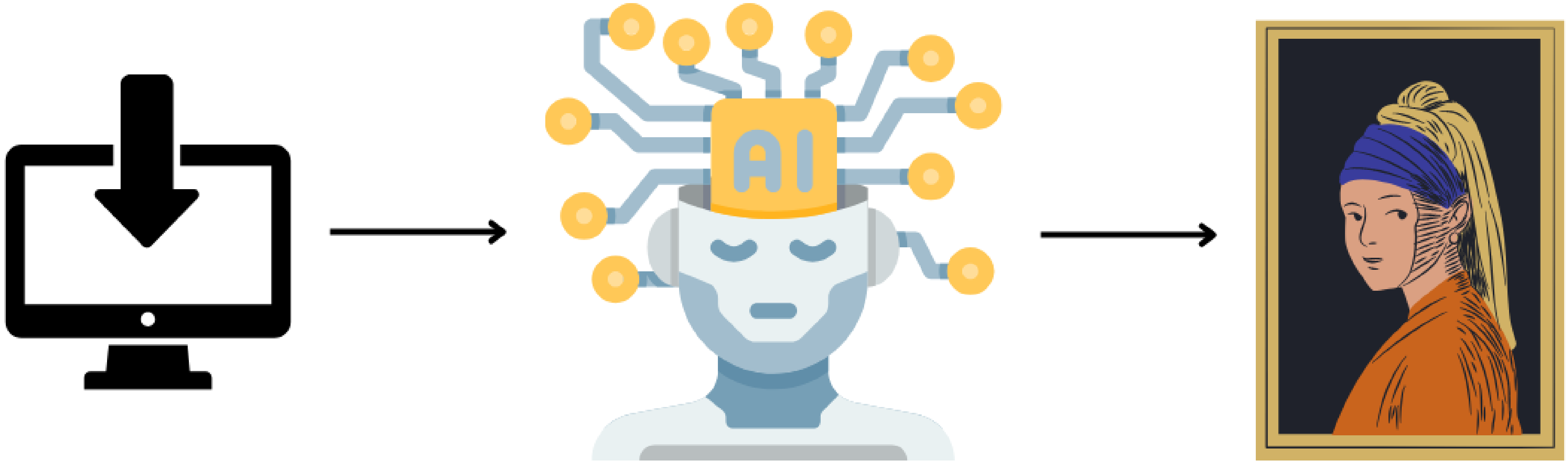


“Immature poets imitate; mature poets steal [...]. The good poet welds his theft into a whole of feeling which is unique, utterly different than that from which it is torn.”

- T.S. Eliot

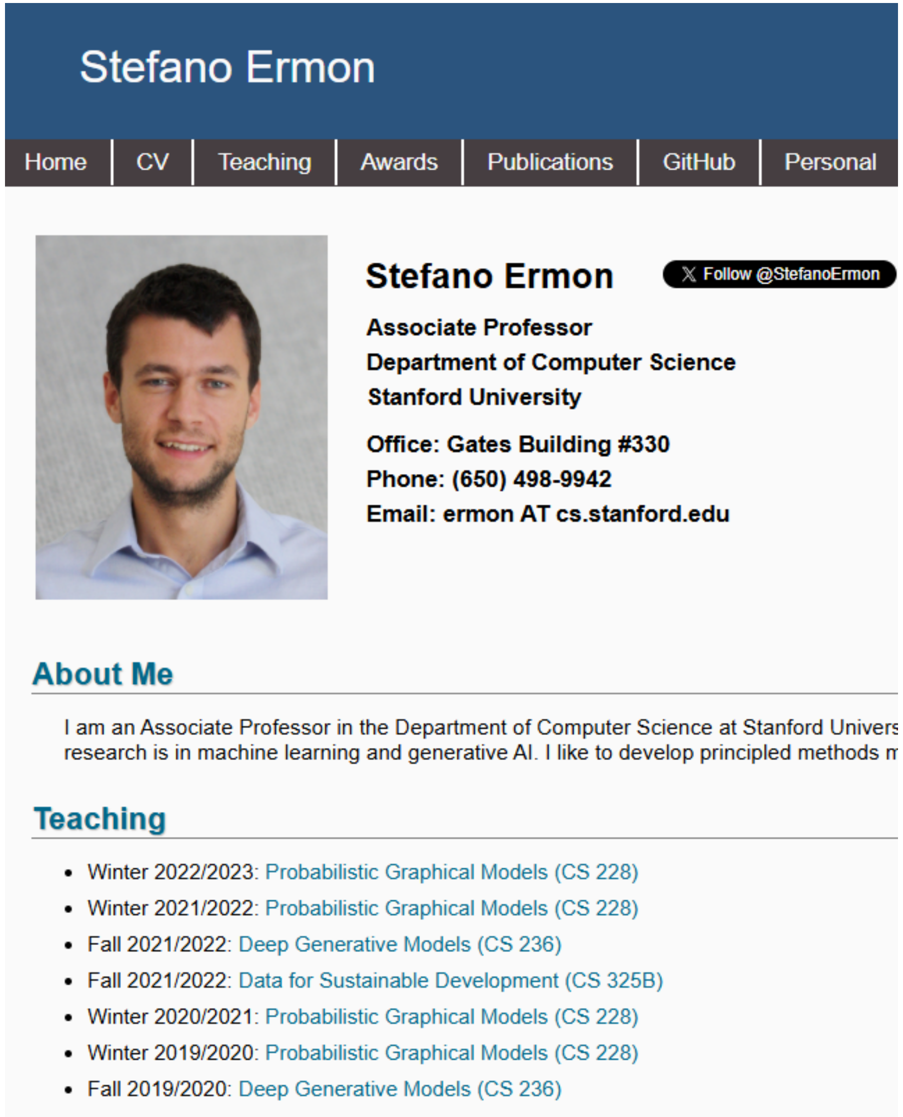
Rabbit Hole
Progress Tracking

Why?



Input Generative Model Output

Why? Philosophical Reason


Stefano Ermon

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About Me

I am an Associate Professor in the Department of Computer Science at Stanford Univers
research is in machine learning and generative AI. I like to develop principled methods m

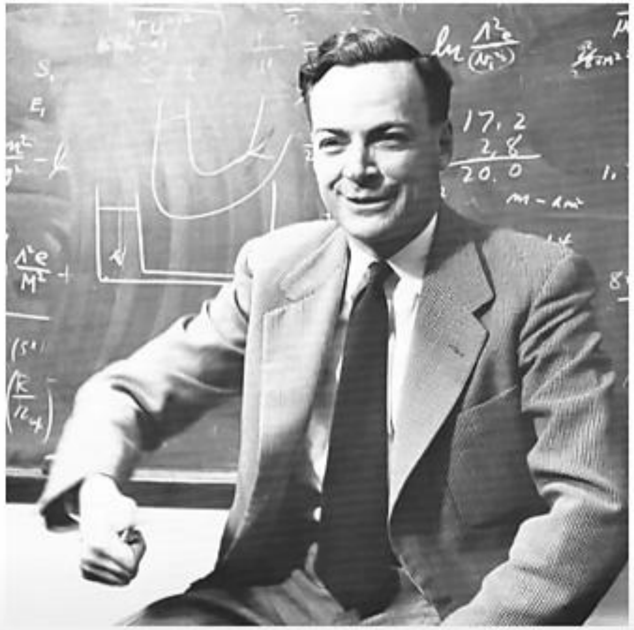
Teaching

- Winter 2022/2023: Probabilistic Graphical Models (CS 228)
- Winter 2021/2022: Probabilistic Graphical Models (CS 228)
- Fall 2021/2022: Deep Generative Models (CS 236)
- Fall 2021/2022: Data for Sustainable Development (CS 325B)
- Winter 2020/2021: Probabilistic Graphical Models (CS 228)
- Winter 2019/2020: Probabilistic Graphical Models (CS 228)
- Fall 2019/2020: Deep Generative Models (CS 236)

Deep Generative Models
Stanford - CS236

Why? Philosophical Reason

Introduction

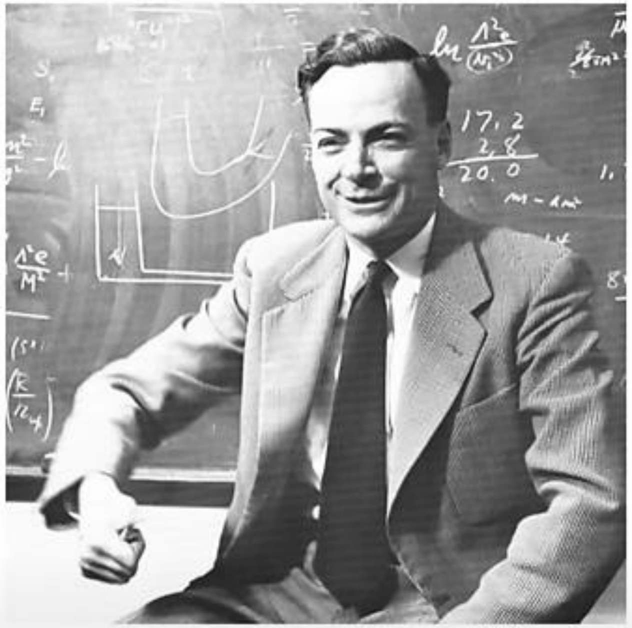


Richard Feynman: "*What I cannot create, I do not understand*"

Stanford

Why? Philosophical Reason

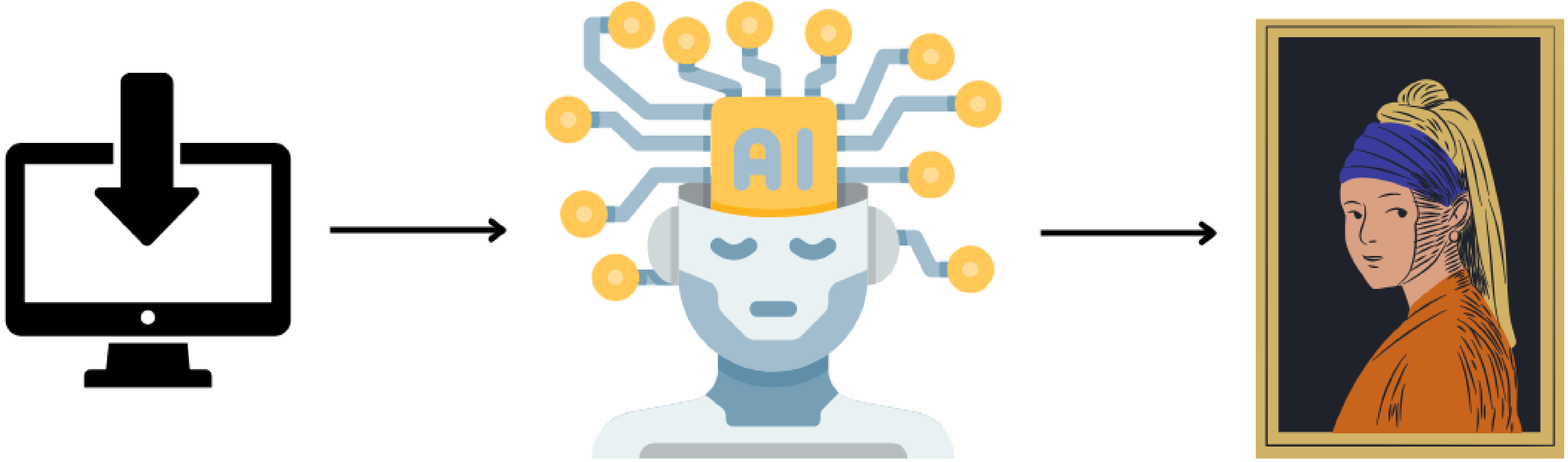
Introduction



Richard Feynman: "*What I cannot create, I do not understand*"

Generative modeling: "*What I understand, I can **create***"

Why? Practical Applications

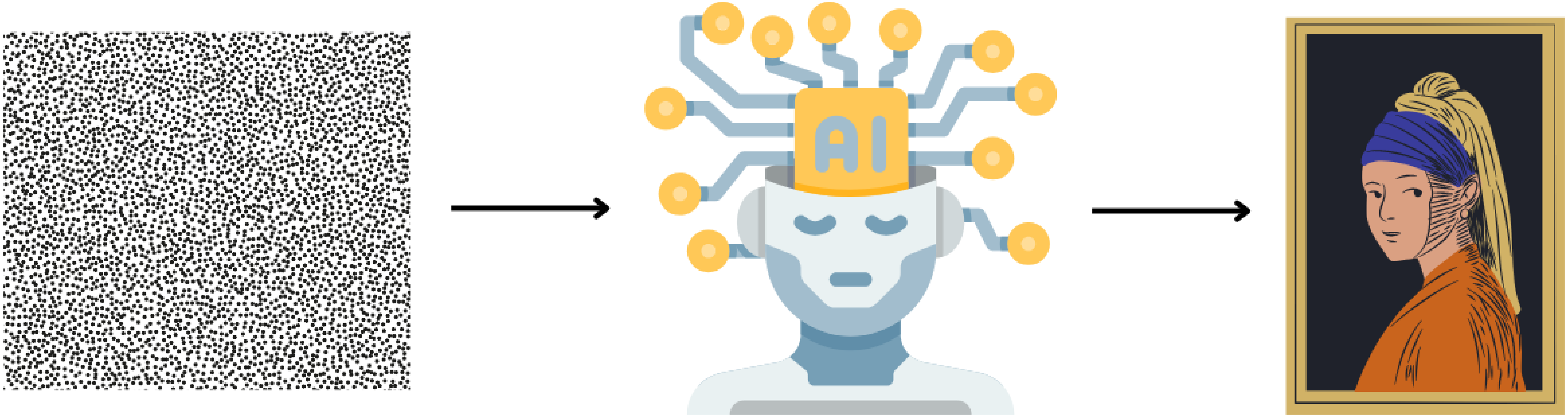


Input Generative Model Output

Why? Practical Applications

- Many problems can be formulated as image generation.
- Based on the input the approach can be:
    1. Noise-to-Image
    2. Text-to-Image
    3. Image-to-Image
    4. Text and Image-to-Image

Noise-to-Image Generation



Noise

Generative Model

Output

Noise-to-Image Generation

Example: https://thispersondoesnotexist.com/

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Image-to-Image Generation

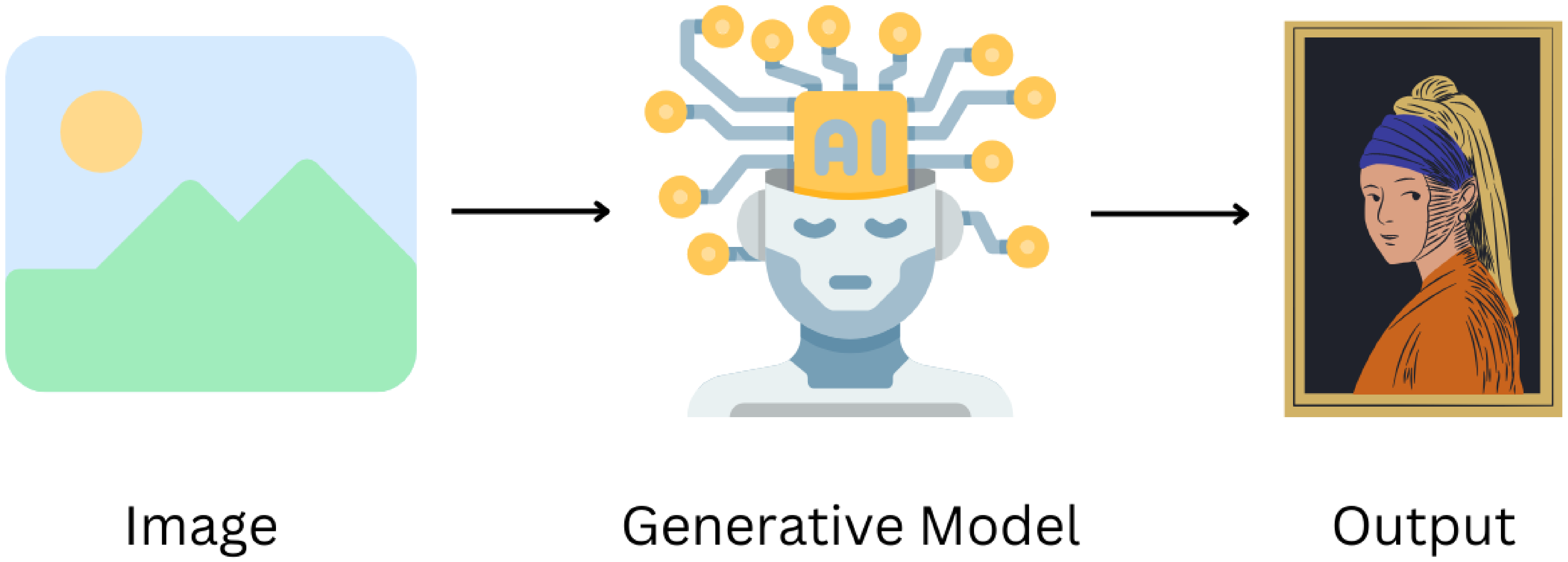
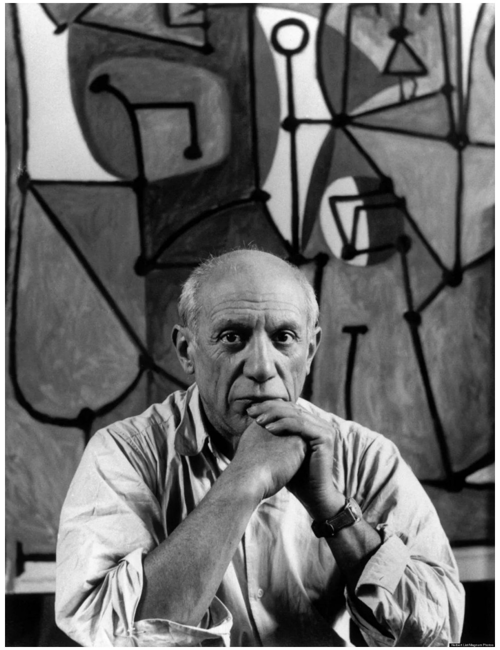


Image-to-Image Generation

Example: Style Transfer - https://reiinakano.com/arbitrary-image-stylization-tfjs/



+

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Input Image

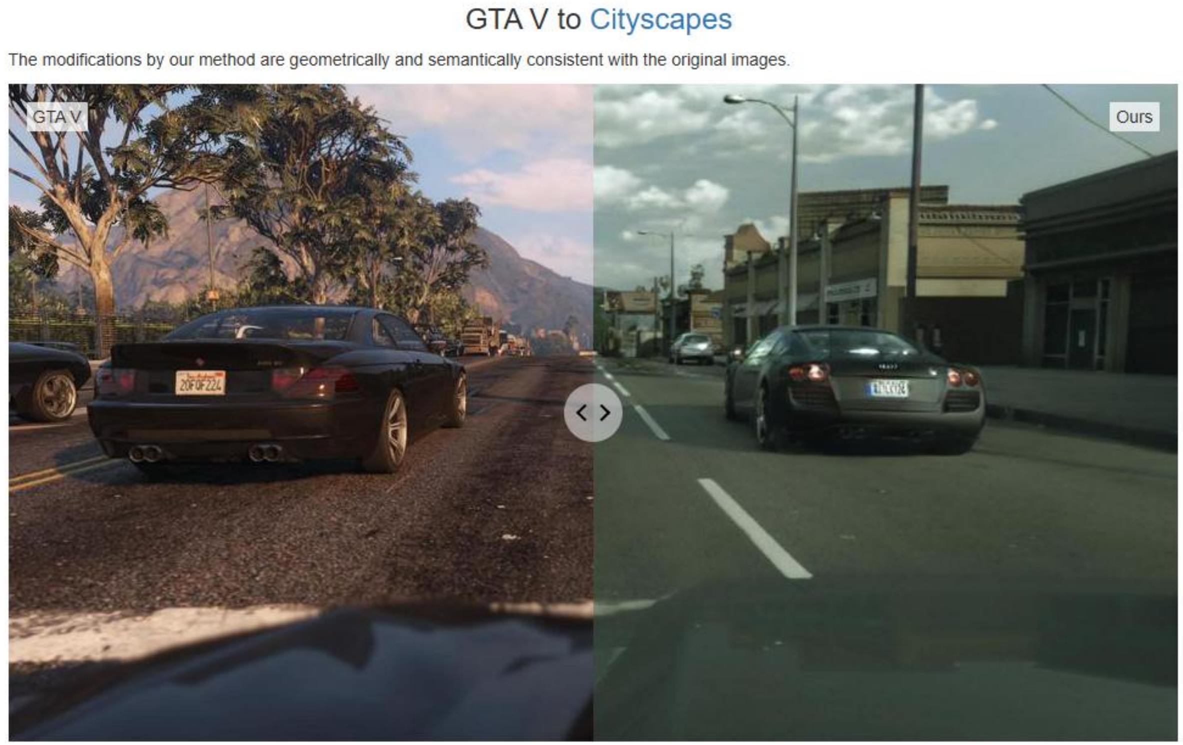
Style Image

Result

Image-to-Image Generation

Application: Enhancing Photorealism Enhancement [1]

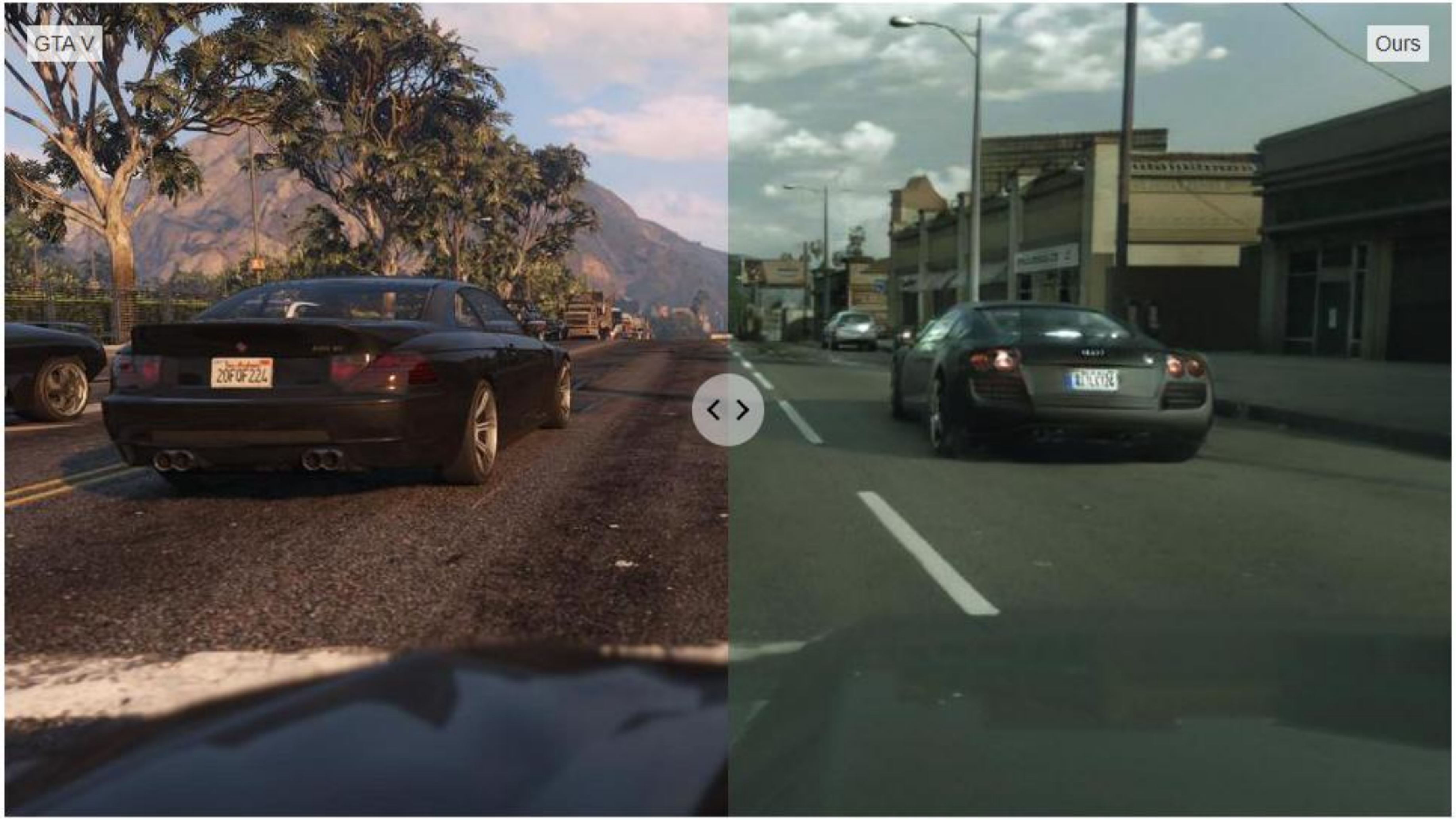
https://isl-org.github.io/PhotorealismEnhancement/



[1] Richter, S.R., AlHaija, H.A. and Koltun, V., 2022. Enhancing photorealism enhancement. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *45*(2), pp.1700-1715.

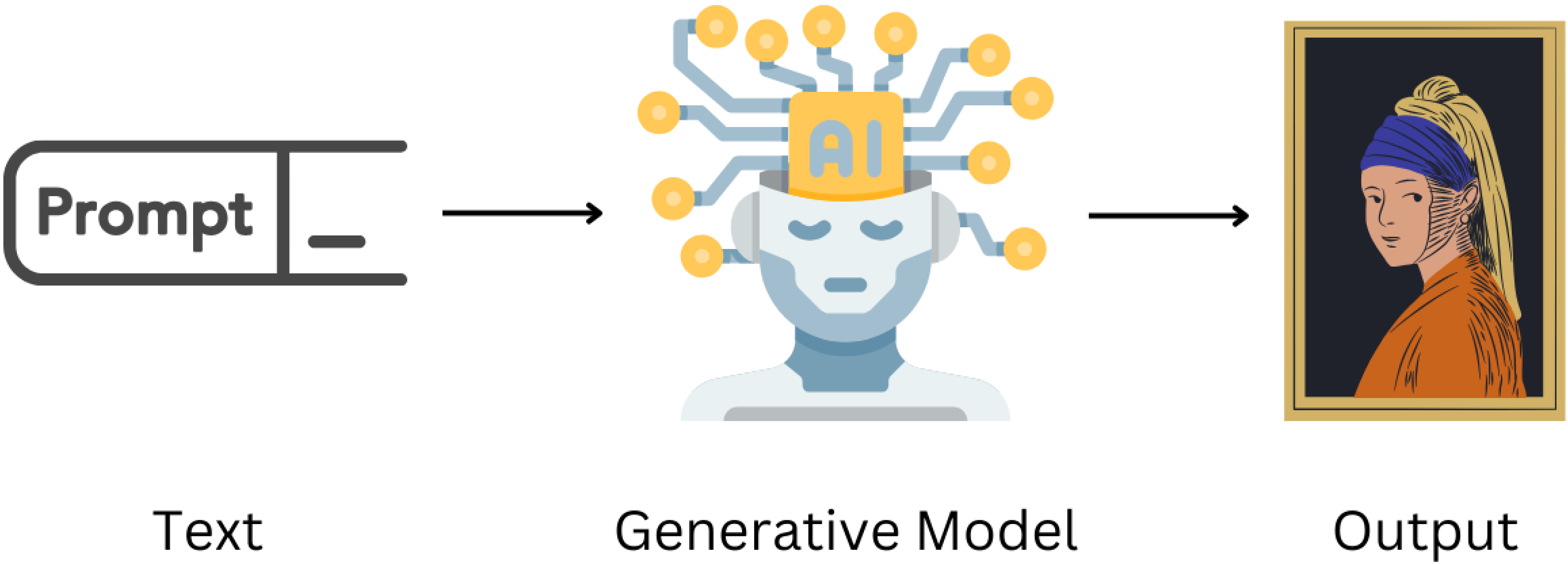
GTA V to Cityscapes

The modifications by our method are geometrically and semantically consistent with the original images.



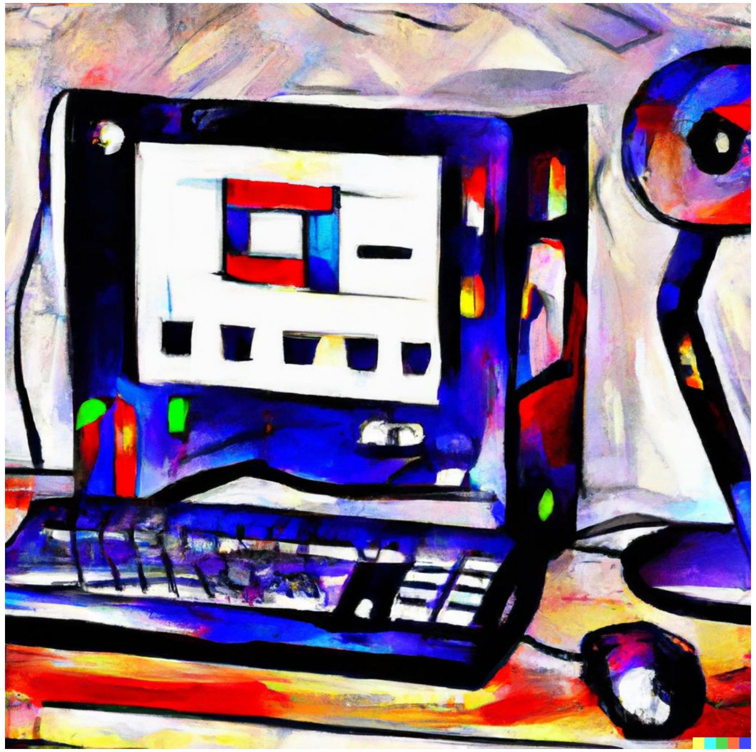
https://isl-org.github.io/PhotorealismEnhancement/

Text-to-Image Generation



Text-to-Image Generation

Example: DALL-E 2 https://openai.com/index/dall-e-2/



Prompt: A painting of a computer in the style of Pablo Picasso

Text-to-Image Generation

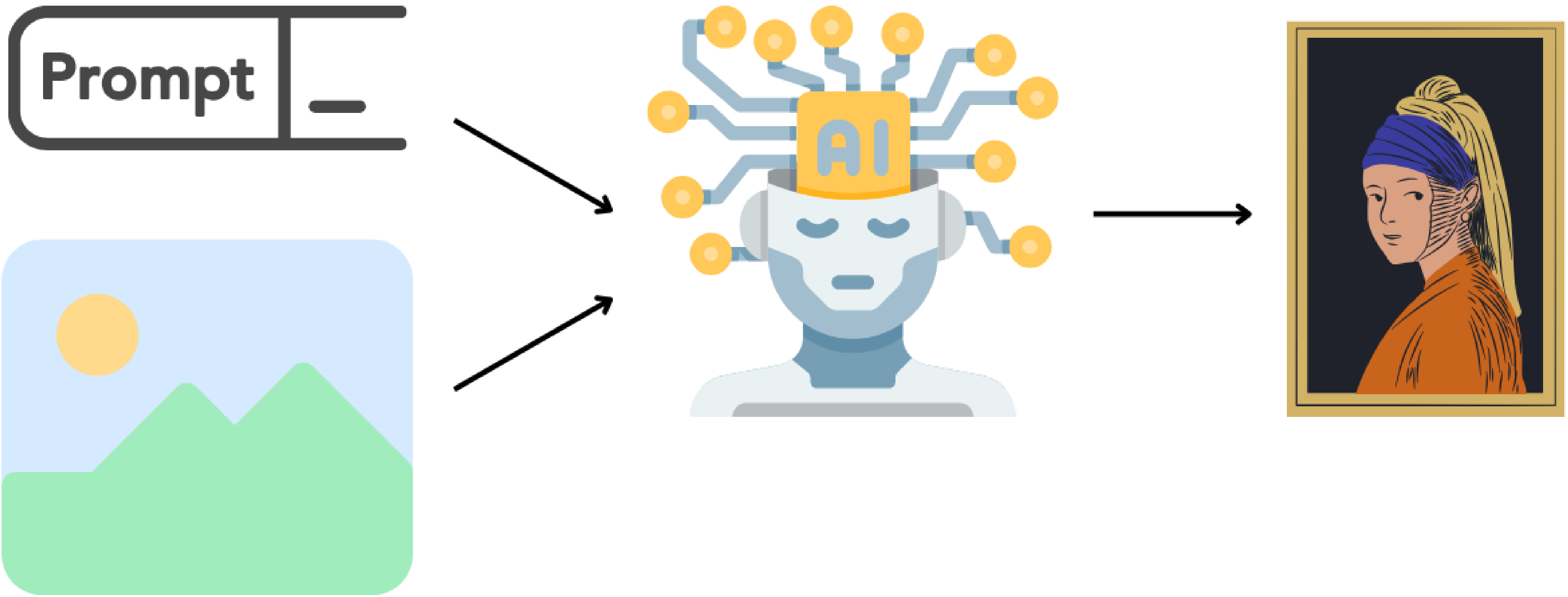
Example: DALL-E 2 https://openai.com/index/dall-e-2/

A hyper-realistic image of a dog quartet playing jazz instruments in a moody club during the night

Generate

A hyper-realistic image of a dog quartet playing jazz instruments in a moody club during the night

Multimodal Image Generation

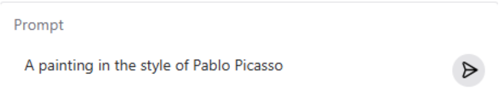


Text & Image Generative Model Output

Multimodal Image Generation

Example: ControlNet https://stablediffusion-online.com/controlnet.php

"A painting in the style of Pablo Picasso"



Input

Result

Why? Practical Applications

- Applications:

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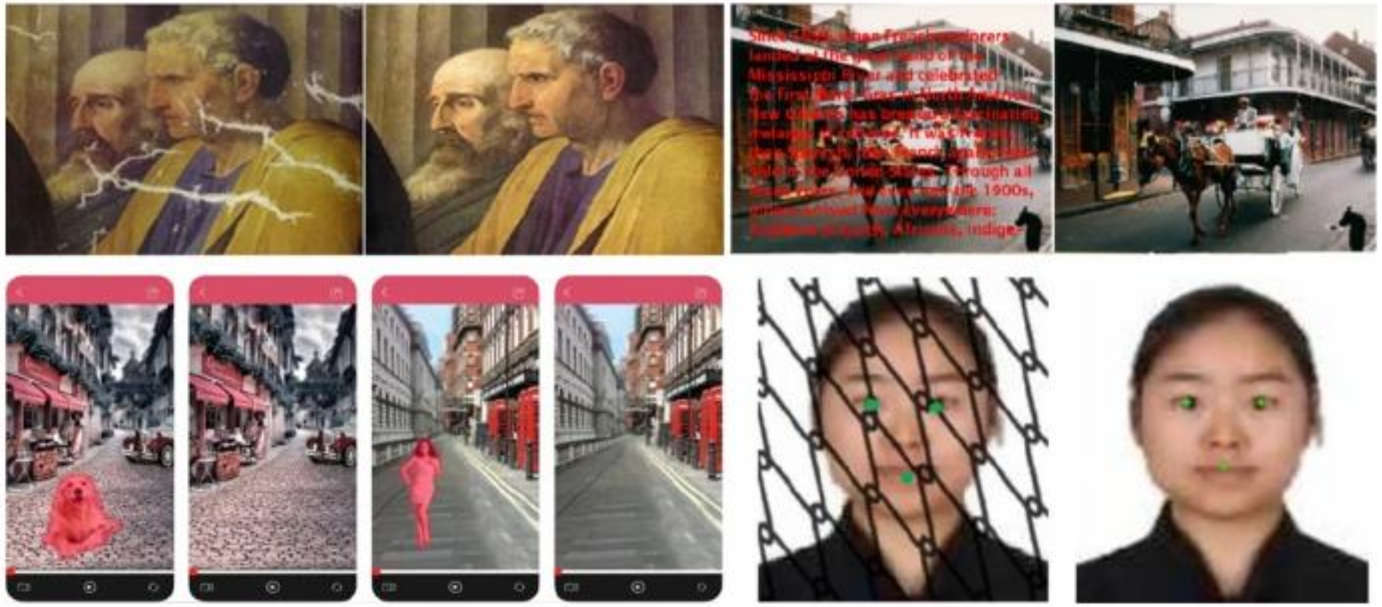
Why? Practical Reason

- Applications:
  1. Image Generation for Dataset Extension
  2. Creativity Boosting
  3. Image Inpainting

Inpainting



Application examples of inpainting techniques: photo restoration (top left: image from (Bertalmio et al., 2000)), text removal (top right: image from (Bertalmio et al., 2000)), undesired target removal (bottom left: image from (Chen, 2018)), and face verification (bottom right: image from (Zhang et al., 2018c))

[2] Quan, Weize, et al. "Deep learning-based image and video inpainting: A survey." International Journal of Computer Vision 132.7 (2024): 2367-2400.

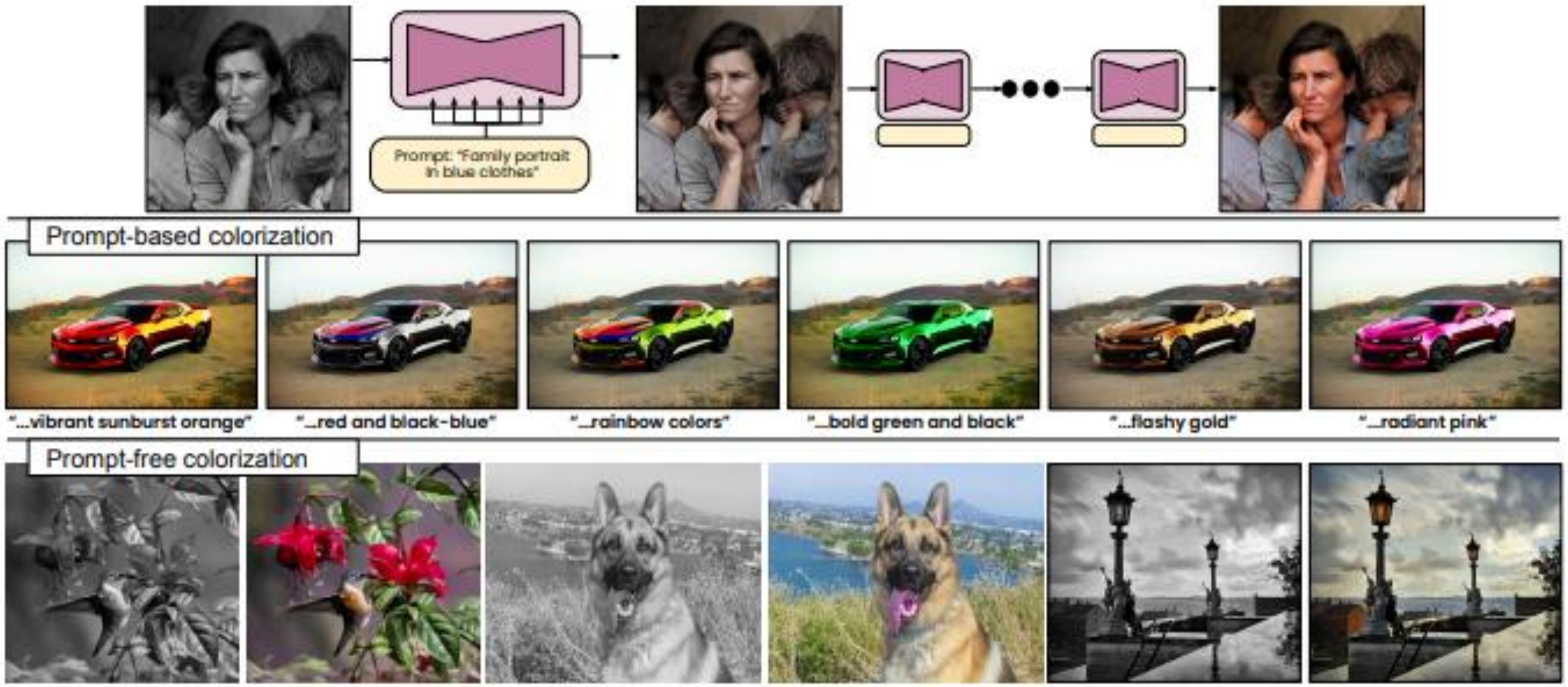
Inpainting

[3] Corneanu, Ciprian, Raghudeep Gadde, and Aleix M. Martinez. "Latentpaint: Image inpainting in latent space with diffusion models." Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. 2024.

Why? Practical Reason

- Applications:
  1. Image Generation for Dataset Extension
  2. Creativity Boosting
  3. Image Inpainting
  4. Image Colorization

Image Colorization



[4] Zabari, Nir, et al. "Diffusing Colors: Image Colorization with Text Guided Diffusion." SIGGRAPH Asia 2023 Conference Papers. 2023.

Image Colorization

- Interesting Git Repo: https://github.com/ErwannMillon/Color-diffusion

Why? Practical Reason

- Applications:
    1. Image Generation for Dataset Extension
    2. Creativity Boosting
    3. Image Inpainting
    4. Image Colorization
    5. Image Super-Resolution

Image Super-Resolution

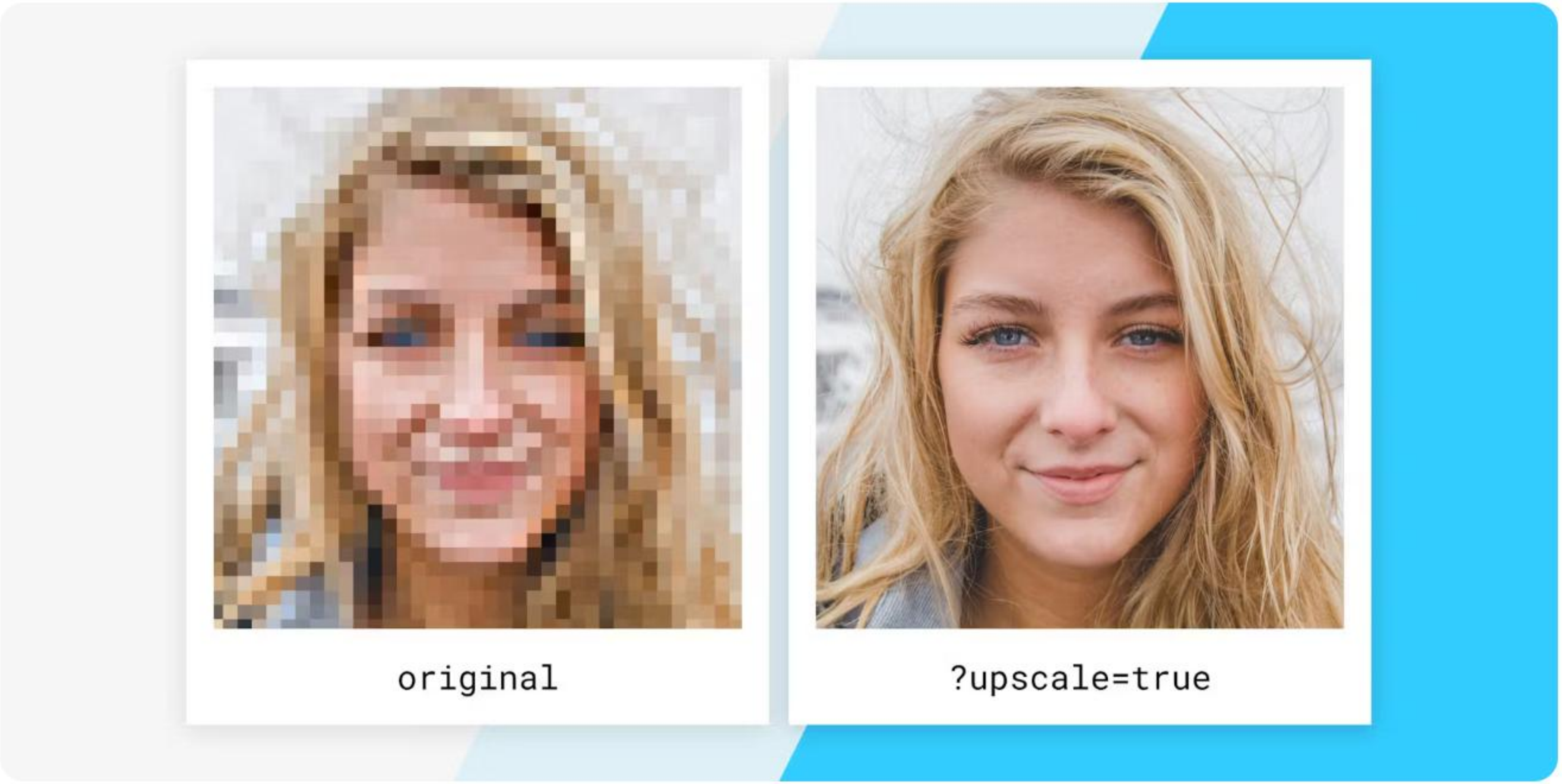


Image from https://www.imgix.com/blog/ai-powered-image-super-resolution

Image Super-Resolution

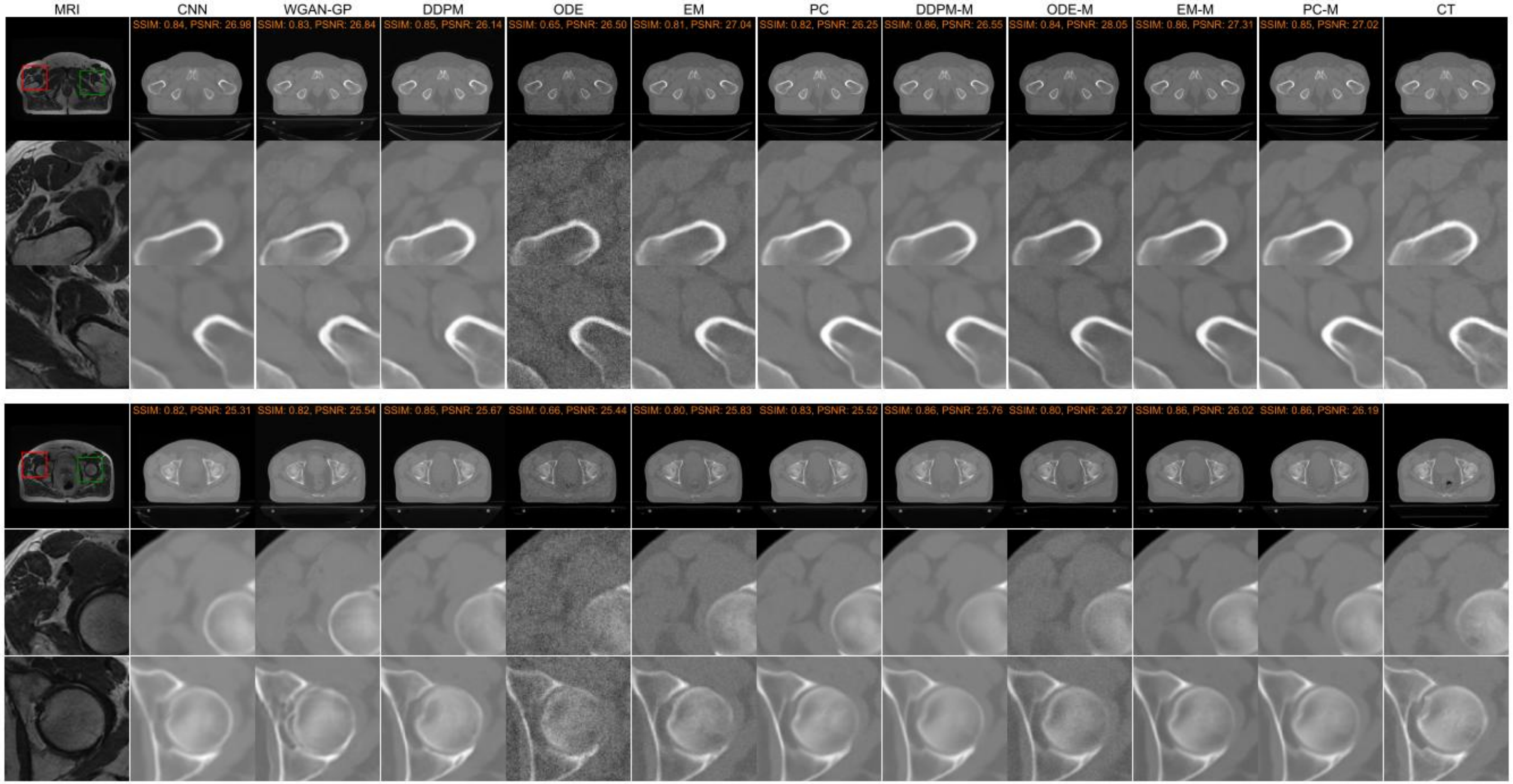
(a) Zoomed LR (b) BSRGAN (c) RealESRGAN (d) SwinIR (e) DASR (f) LDM-15 (g) *ResShift* (h) Ground Truth

[5] Yue, Zongsheng, Jianyi Wang, and Chen Change Loy. "Resshift: Efficient diffusion model for image super-resolution by residual shifting." Advances in Neural Information Processing Systems 36 (2024).

Why? Practical Reason

- Some Applications:
  1. Image Generation for Dataset Extension
  2. Creativity Boosting
  3. Image Inpainting
  4. Image Colorization
  5. Image Super-Resolution
  6. Medical Image Conversion

Transform MRI into CT



[6] Lyu, Qing, and Ge Wang. "Conversion between CT and MRI images using diffusion and score-matching models." arXiv preprint arXiv:2209.12104 (2022).

Why Stable Diffusion?

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Why Stable Diffusion?

- Generative Models:
  1. Variational Autoencoders (2013) [7]
  2. Generative Adversarial Networks (2014) [8]
  3. Stable Diffusion (2020) [9]

[7] Kingma, Diederik P. "Auto-encoding variational bayes." arXiv preprint arXiv:1312.6114 (2013).

[8] Goodfellow, Ian, et al. "Generative adversarial nets." Advances in neural information processing systems 27 (2014).

[9] Ho, Jonathan, Ajay Jain, and Pieter Abbeel. "Denoising diffusion probabilistic models." Advances in neural information processing systems 33 (2020): 6840-6851.

Why Stable Diffusion?

- Generative Models:
  1. Variational Autoencoders (2013) [7]
  2. Generative Adversarial Networks (2014) [8]
  3. Denoising Diffusion Probabilistic Models (DDPMs) (2020) [9]
  4. Deep Unsupervised Learning using Nonequilibrium Thermodynamics (2015) [10]

[7] Kingma, Diederik P. "Auto-encoding variational bayes." arXiv preprint arXiv:1312.6114 (2013).

[8] Goodfellow, Ian, et al. "Generative adversarial nets." Advances in neural information processing systems 27 (2014).

[9] Ho, Jonathan, Ajay Jain, and Pieter Abbeel. "Denoising diffusion probabilistic models." Advances in neural information processing systems 33 (2020): 6840-6851.

[10] Sohl-Dickstein, Jascha, et al. "Deep unsupervised learning using nonequilibrium thermodynamics." International conference on machine learning. PMLR, 2015.

Why Stable Diffusion?

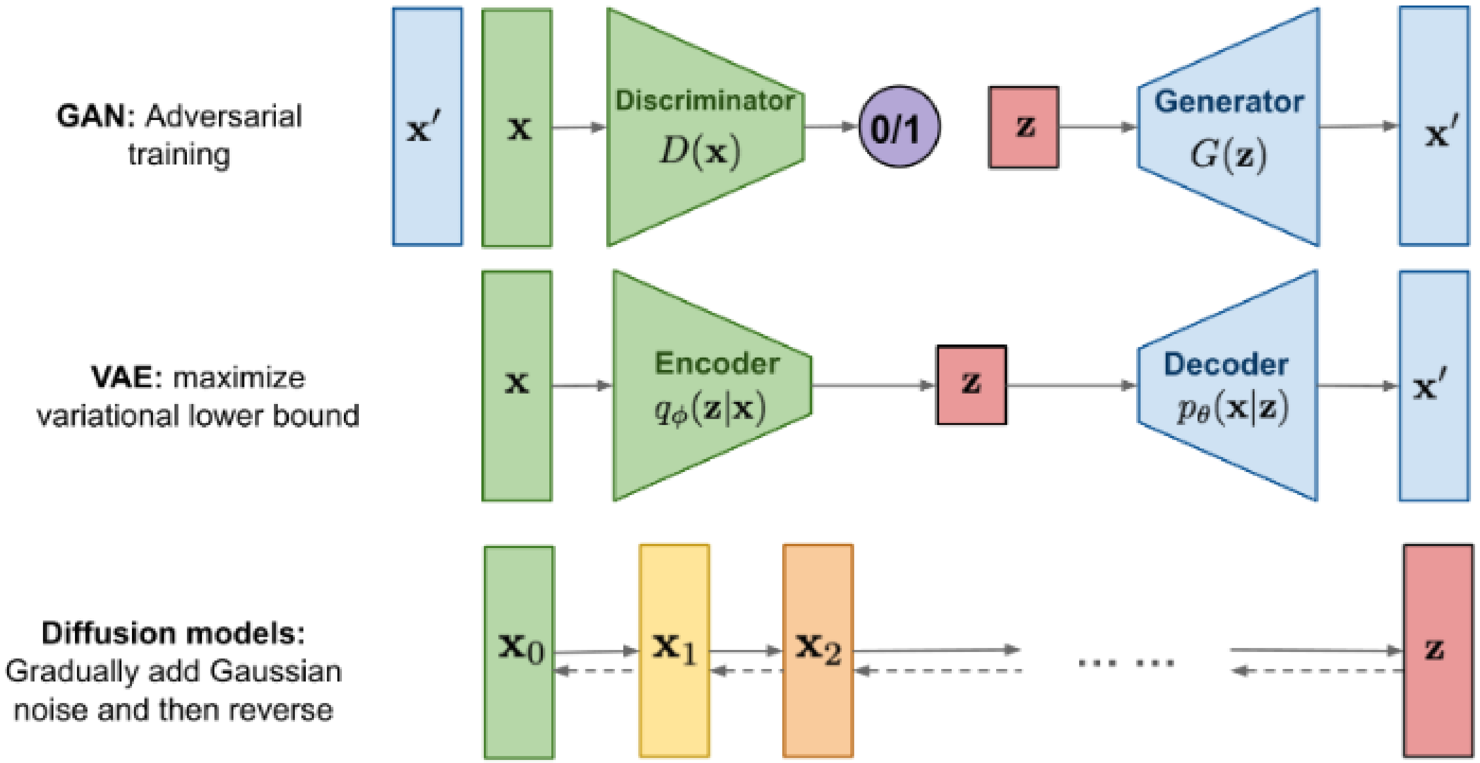

Image from https://lilianweng.github.io/posts/2021-07-11-diffusion-models/

Intuition Check

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Why Stable Diffusion?

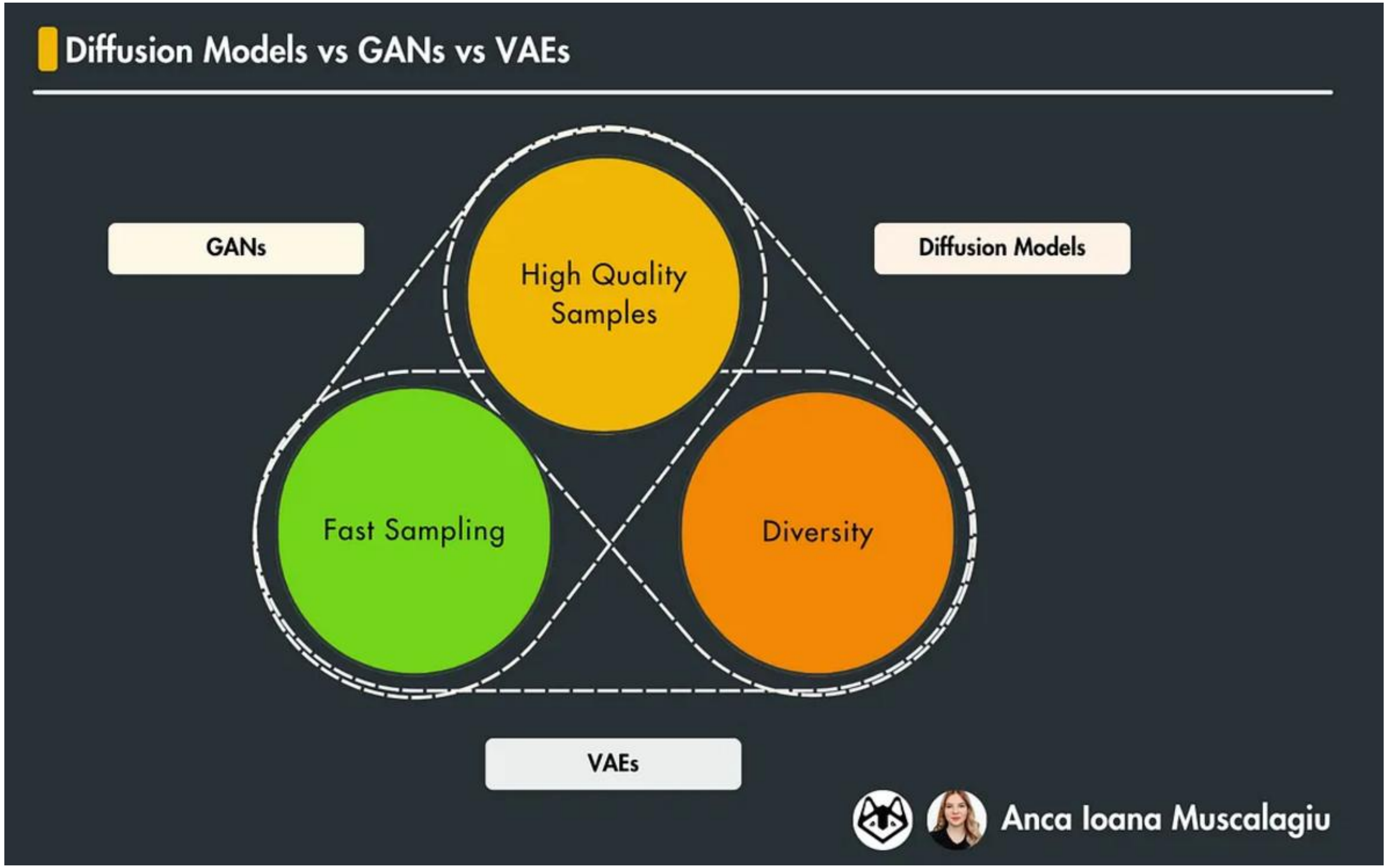


Image from https://medium.com/decodingml/the-abcs-of-diffusion-models-51902a331068

Why Stable Diffusion?

- High-quality samples
- Diverse sample
- **Cost:** Slow Sampling

What is diffusion?

dexonline Despre noi Implică-te Resurse Anonim

difuziune caută opțiuni

Forma **difuzie** este o variantă a lui **difuziune**.

sinteză (1) definiții (17) declinări

Aceste definiții sunt compilate de echipa dexonline. Definițiile originale se află pe fila definiții. Puteți reordona filele pe pagina de preferințe.

ascunde

arată:
- sensuri secundare
- expresii
- exemple
- surse

difuziune, difuziuni / **difuzie**, difuzii substantiv feminin

1. Împrăștierea în toate direcțiile razelor unui fascicul de lumină, a undelor de radio etc. care trec printr-un mediu translucid sau care se reflectă când întâlnesc o suprafață cu asperități.
   sinonime: propagare răspândire împrăștiere
   1.1. Difuzare de unde radiofonice.
   sinonime: difuzare

2. Pătrunderea moleculelor unui corp în masa altui corp cu care vin în contact.
   2.1. Întrepătrundere a două lichide puse în contact fără a fi agitație.

etimologie:

limba franceză **diffusion**

limba latină **diffusio, -onis**

What is diffusion?

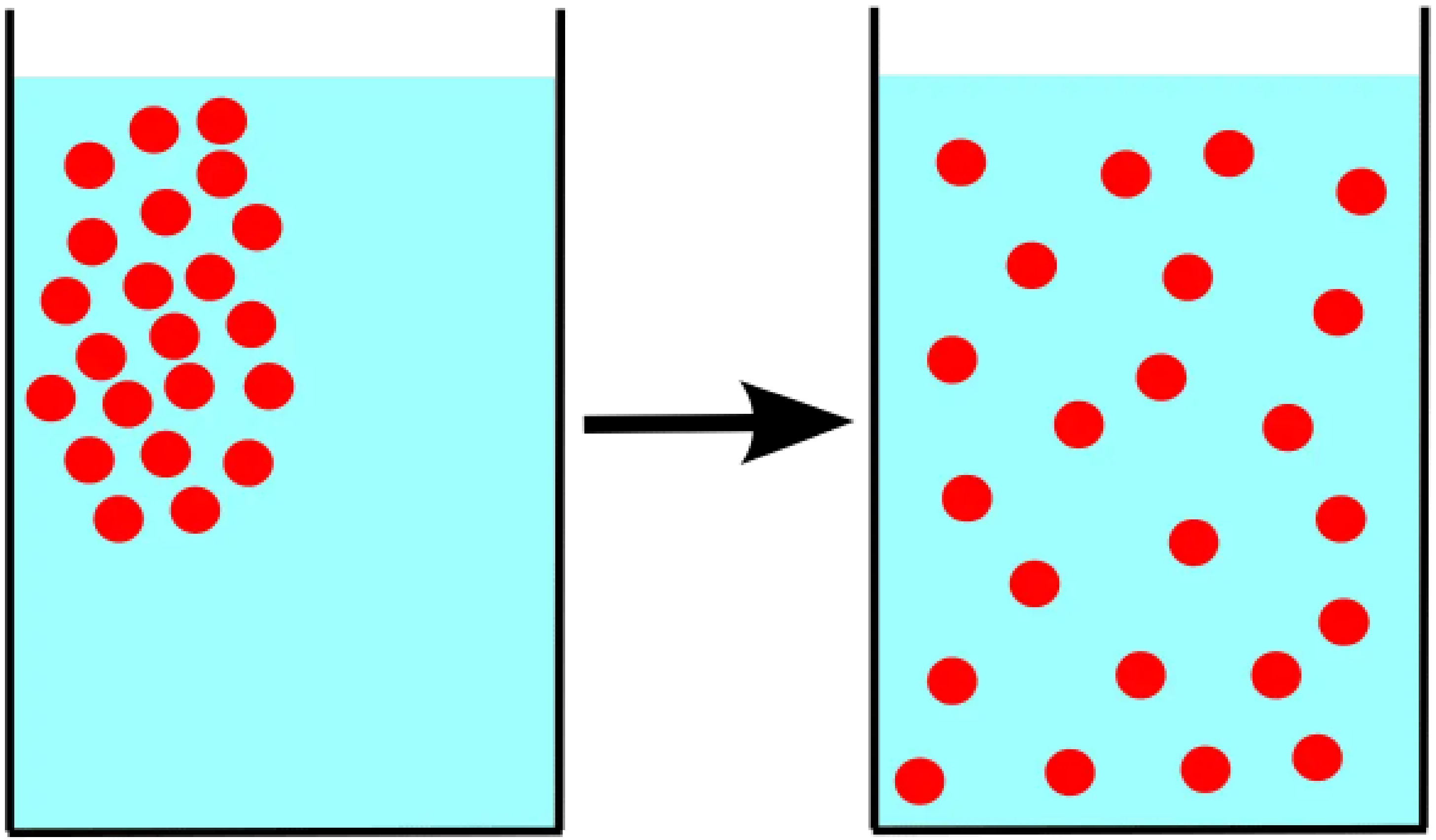


Image from https://biologyjunction.com/osmosis-and-diffusion/

What is diffusion?

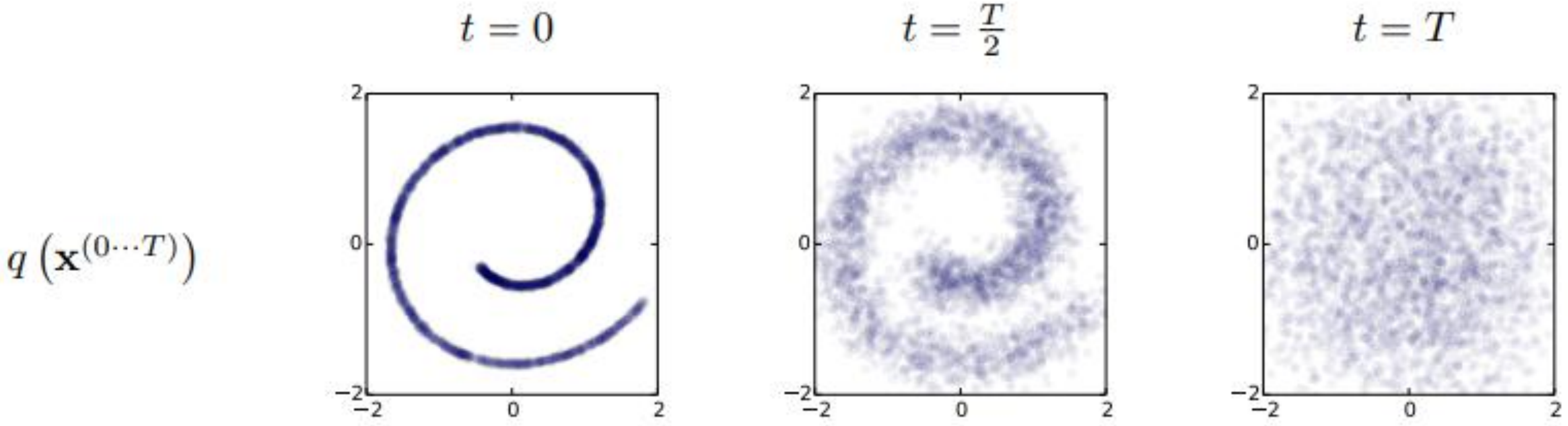


Image from Sohl-Dickstein, Jascha, et al. "Deep unsupervised learning using nonequilibrium thermodynamics." International conference on machine learning. PMLR, 2015.

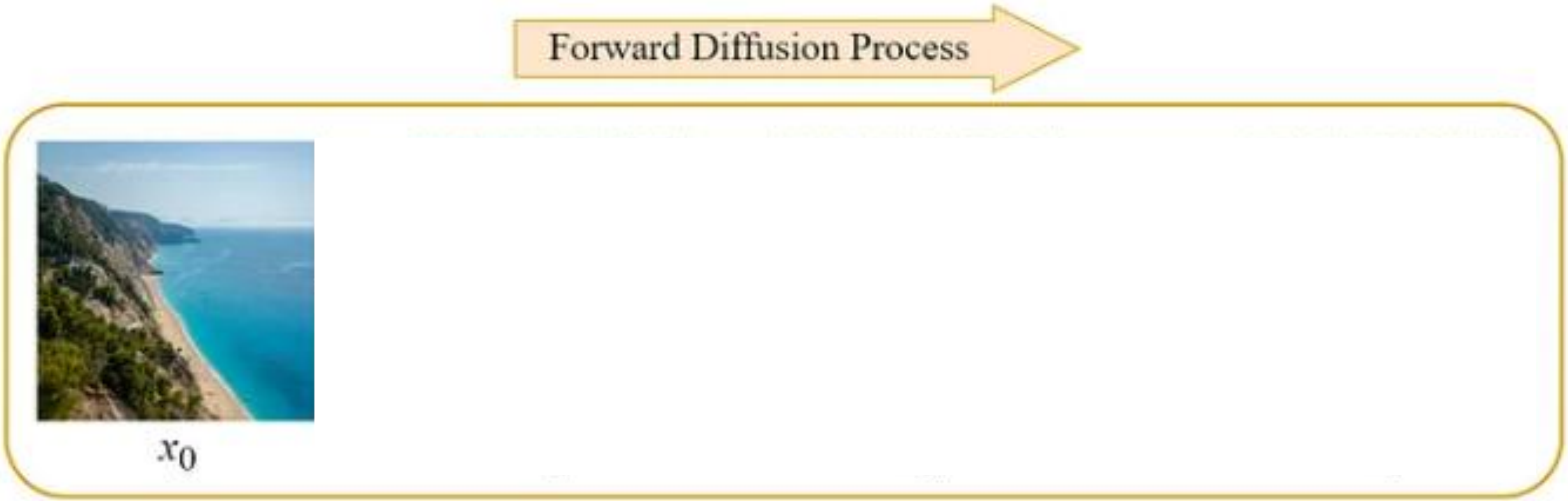
What is diffusion?

Image from https://blog.marvik.ai/

How does it work?

1. Forward Diffusion
2. “Reparameterization Trick”
3. Training Procedure
4. Generation Procedure

Forward Diffusion



Diffusion Markov Chain. Image from https://medium.com/@jaskaranbhatia/summarizing-the-evolution-of-diffusion-models-insights-from-three-research-papers-6889339eba4

Refresher – What is an image?

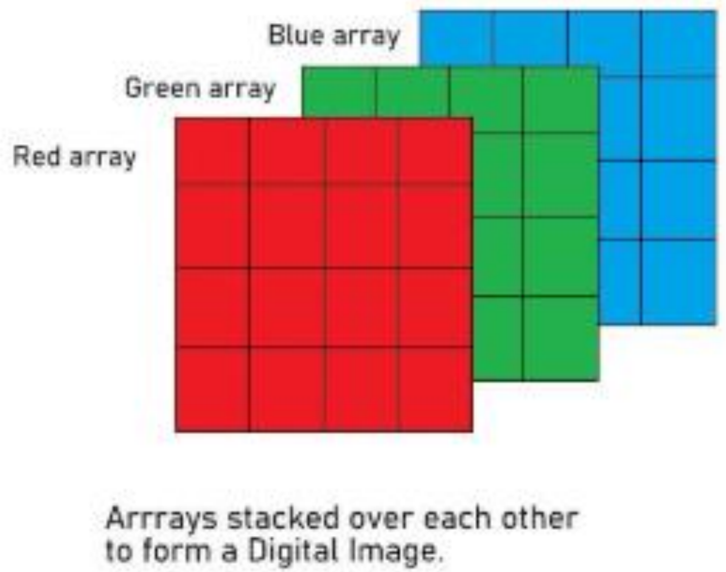


Image from https://www.geeksforgeeks.org/matlab-rgb-image-representation/

Refresher – What is an image?

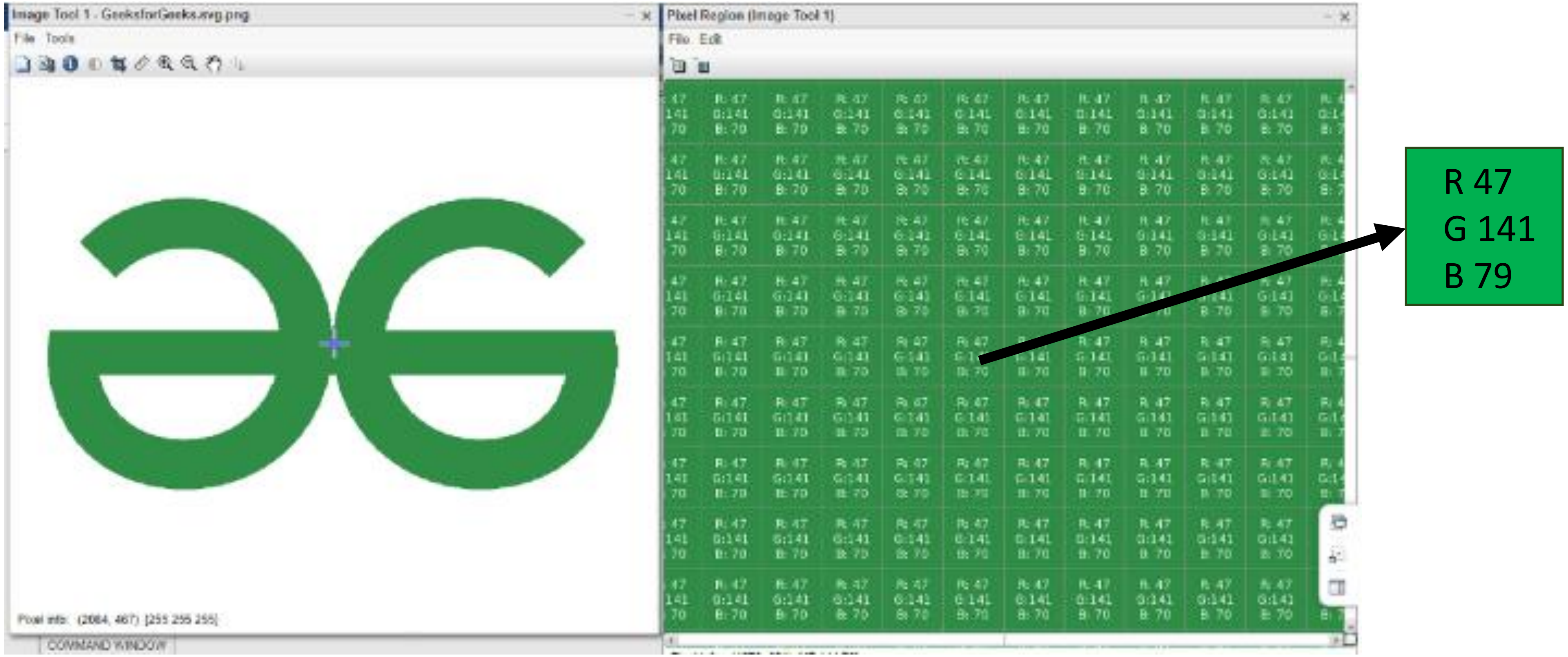


Image from https://www.geeksforgeeks.org/matlab-rgb-image-representation/

Forward Diffusion – Adding noise

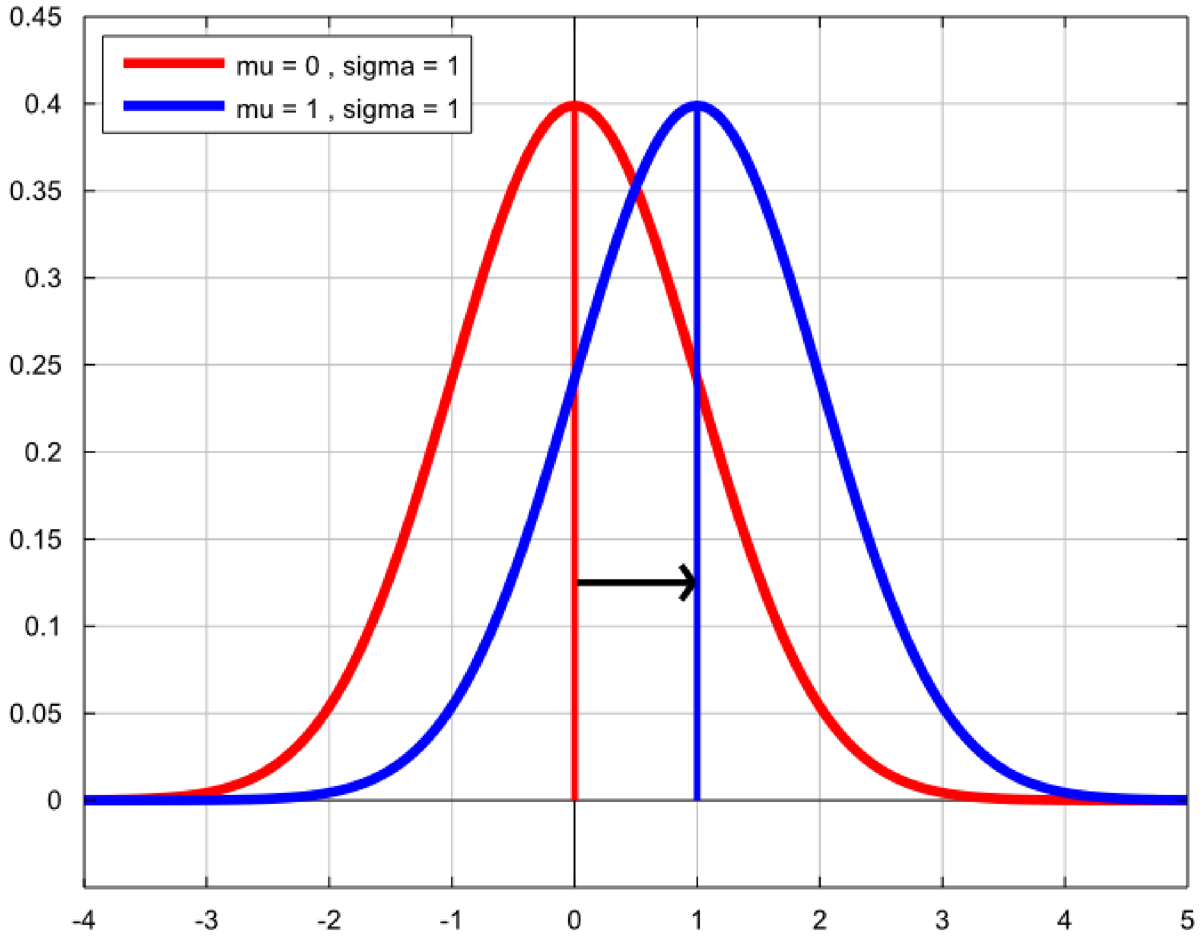


Image from https://www.statlect.com/probability-distributions/normal-distribution

Forward Diffusion – Adding noise

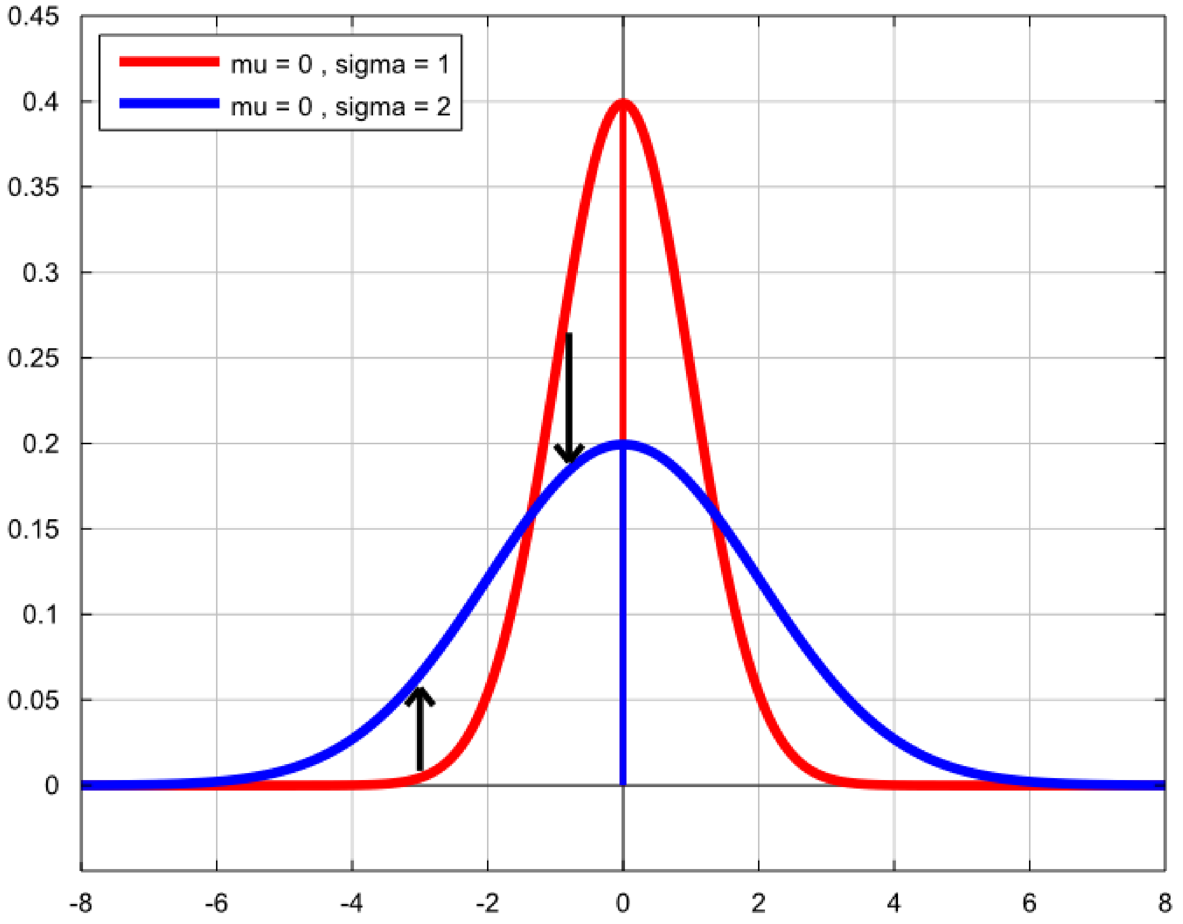


Image from https://www.statlect.com/probability-distributions/normal-distribution

Forward Diffusion – Adding noise

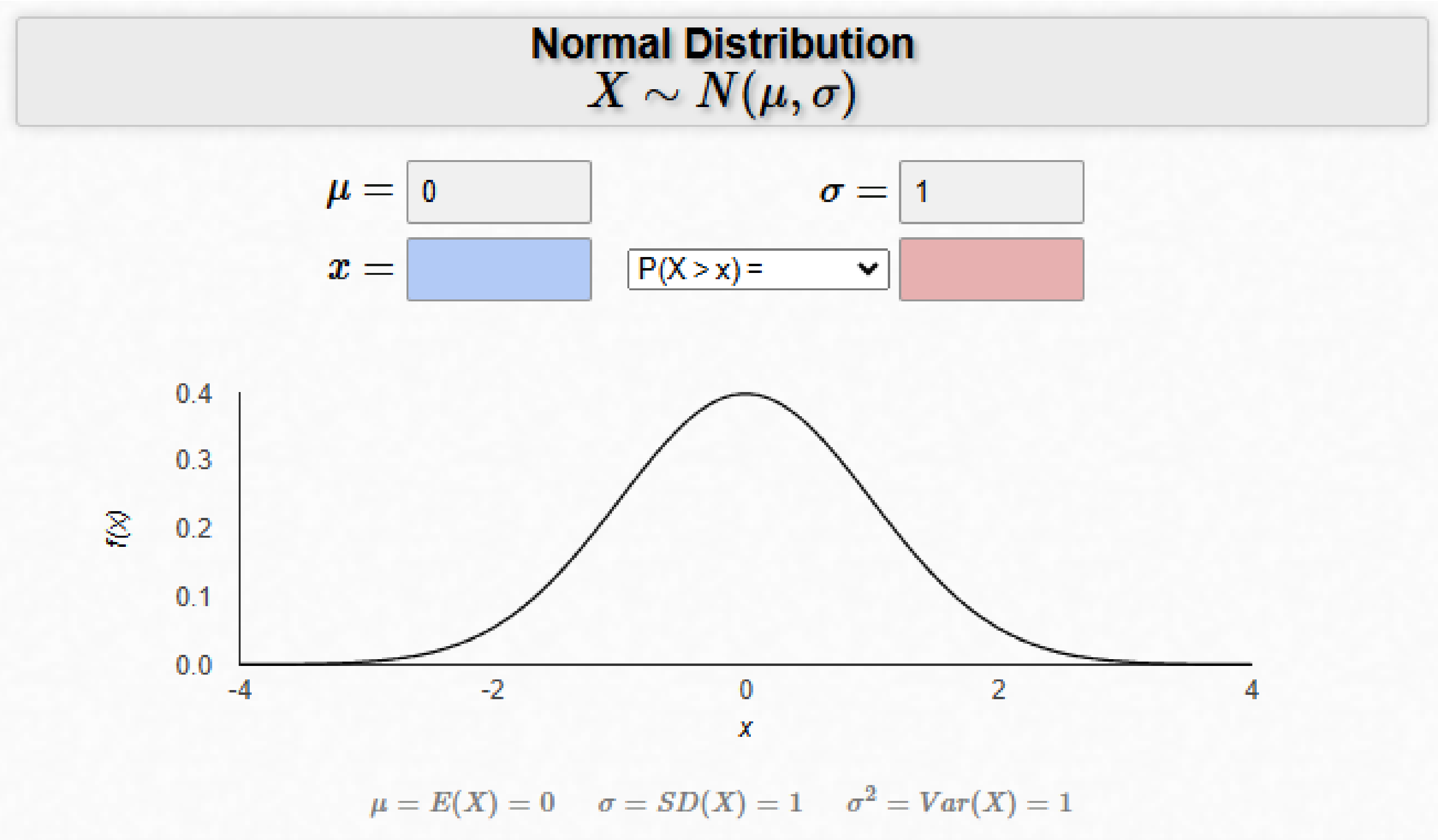


Image from https://homepage.divms.uiowa.edu/~mbognar/applets/normal.html

Forward Diffusion – Adding noise

How does the variance of the Gaussian noise influence the image?
https://explore.albumentations.ai/transform/GaussNoise

Instructions

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Forward Diffusion – Adding noise

(d) Gaussian Noise Mean=0, Variance = 0.002

(e) Gaussian Noise Mean=0, Variance = 0.05

(f) Gaussian Noise Mean=0, Variance = 0.3

Image from https://www.researchgate.net/figure/Gaussian-Noise-with-different-variances-and-corresponding-decrypted-images_fig2_326215771

Forward Diffusion - Scheduling

Image from Nichol, Alexander Quinn, and Prafulla Dhariwal. "Improved denoising diffusion probabilistic models." International conference on machine learning. PMLR, 2021.

Forward Diffusion

Image from https://www.iguazio.com/glossary/diffusion-models/

Reparameterization Trick

Image from https://www.iguazio.com/glossary/diffusion-models/

The Task: Reverse Diffusion

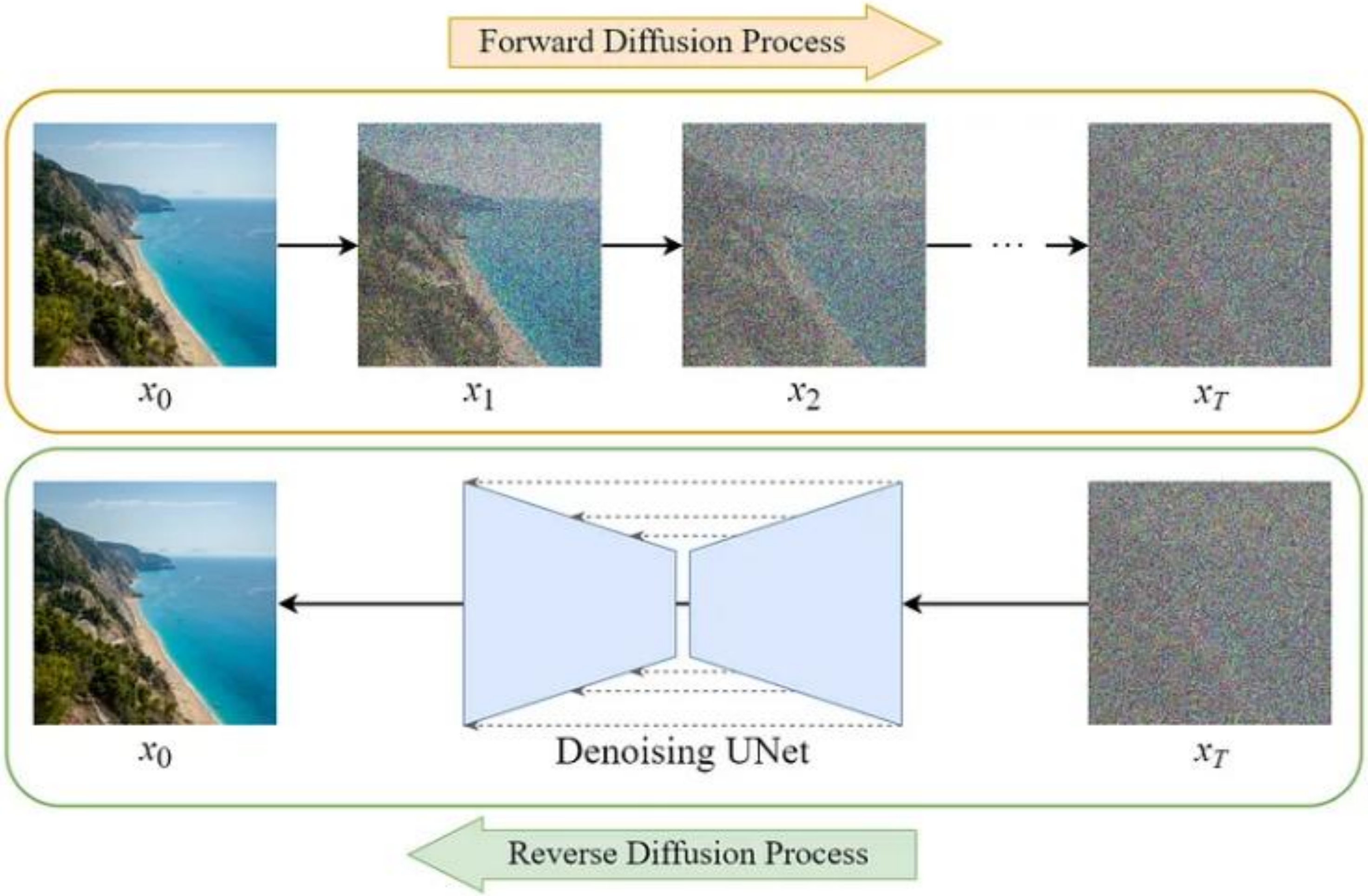
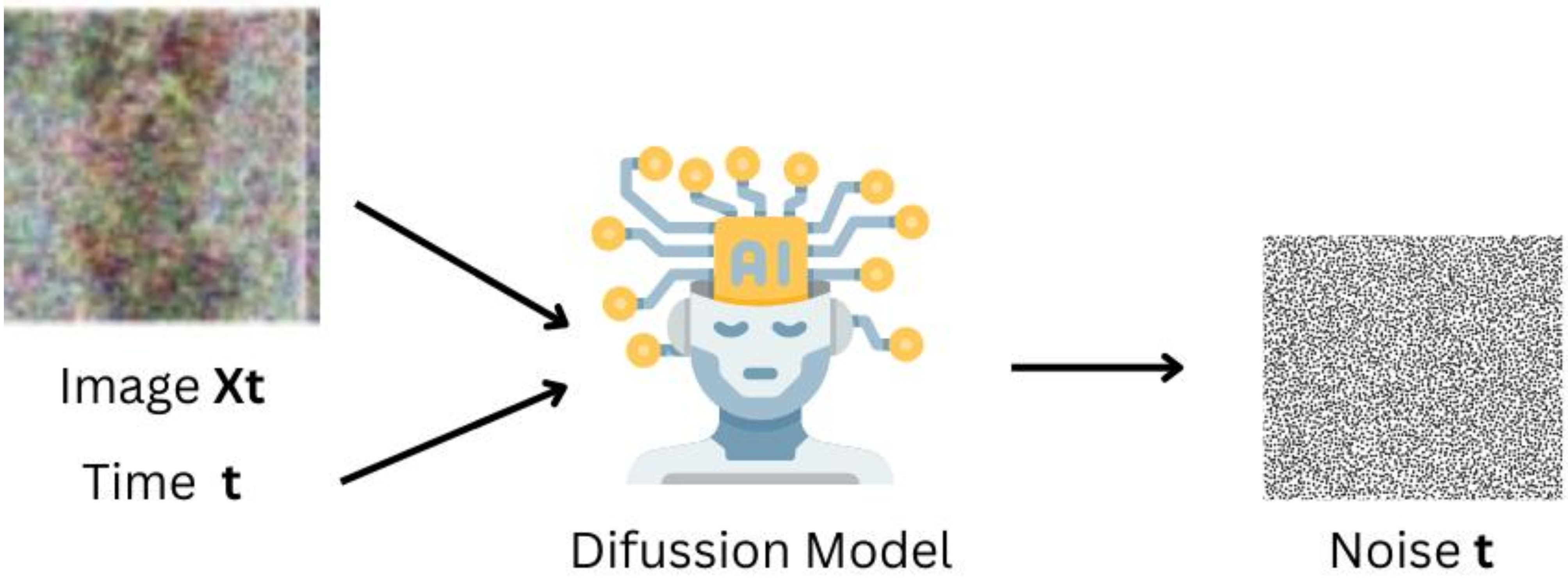
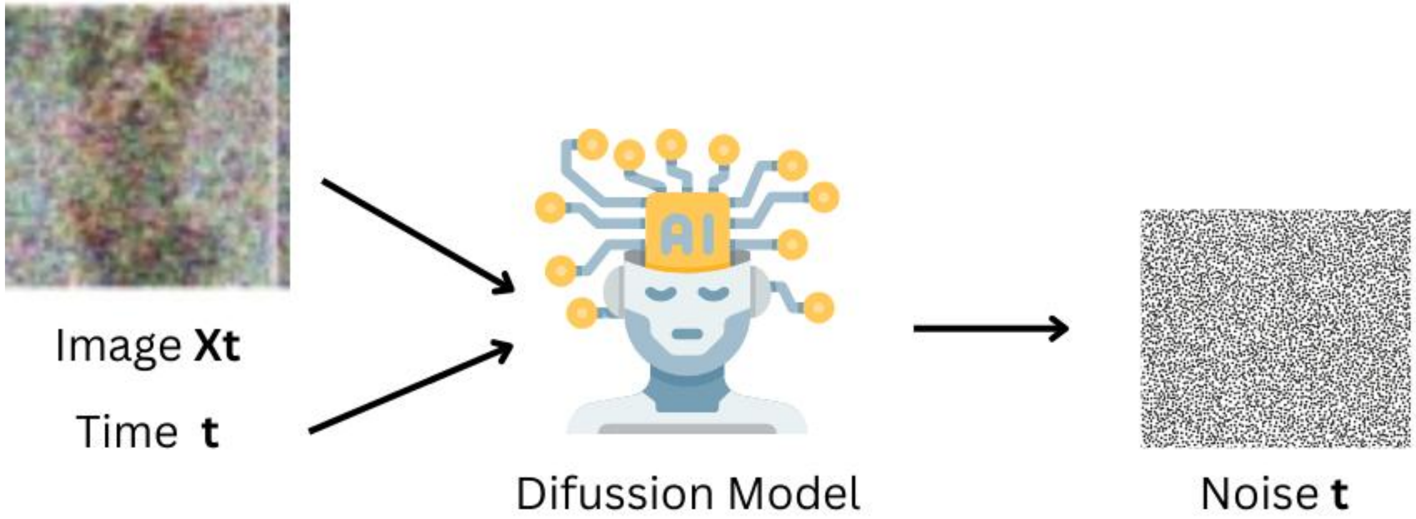


Image from https://medium.com/@jaskaranbhatia/summarizing-the-evolution-of-diffusion-models-insights-from-three-research-papers-6889339eba4

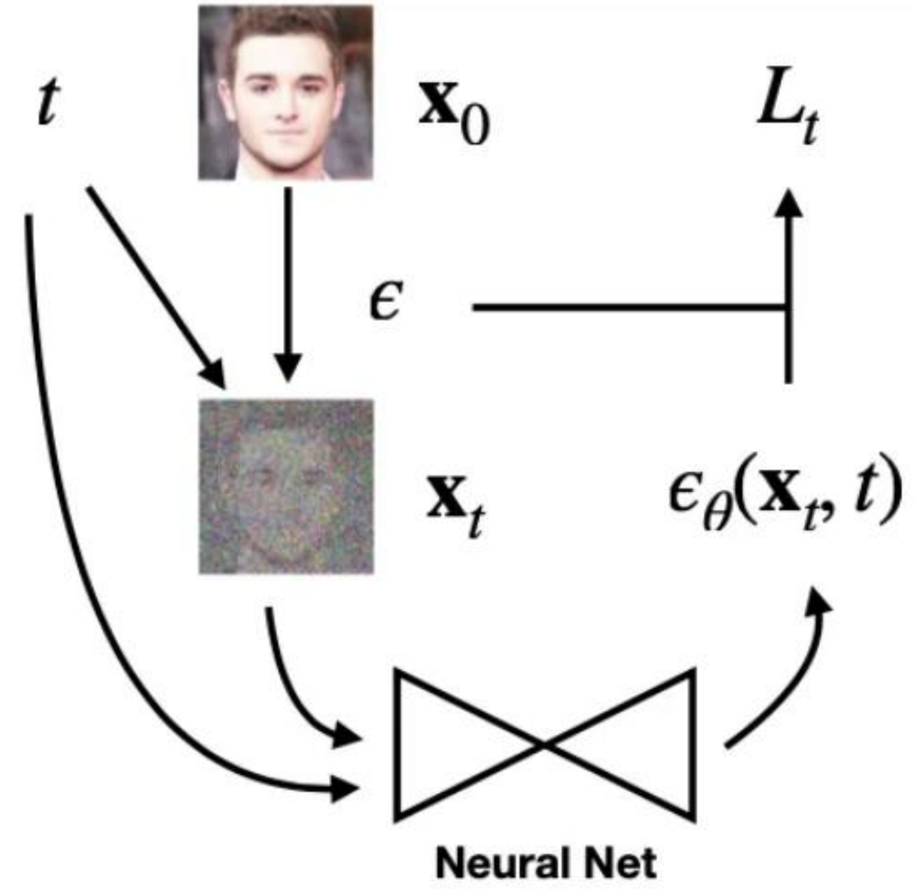
Training Procedure



Training Procedure



Training Procedure



Stable Diffusion Training Process. Image from https://medium.com/@jjkoz762/diffusion-models-2d303735d219

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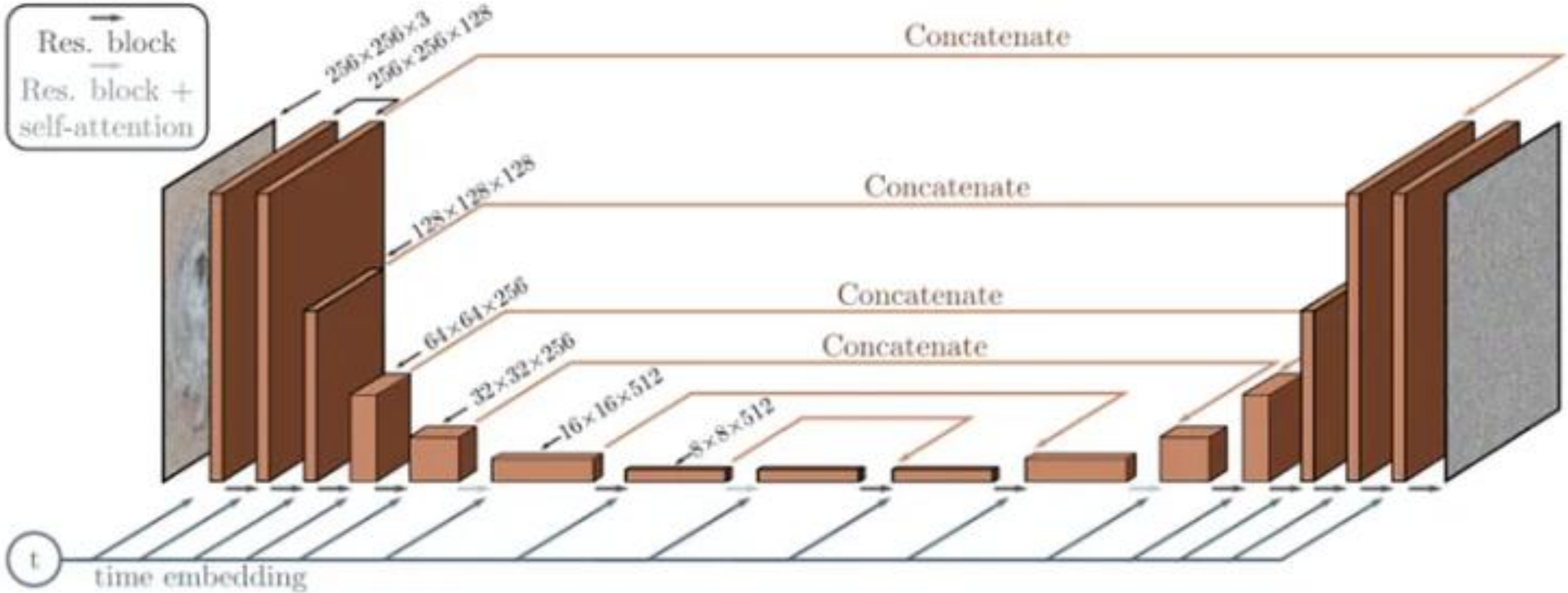
Stable Diffusion – U-Net



U-Net[11].Image from https://medium.com/@jjkoz762/diffusion-models-2d303735d219

[11] Ronneberger, Olaf, Philipp Fischer, and Thomas Brox. "U-net: Convolutional networks for biomedical image segmentation." Medical image computing and computer-assisted intervention–MICCAI 2015: 18th international conference, Munich, Germany, October 5-9, 2015, proceedings, part III 18. Springer International Publishing, 2015.

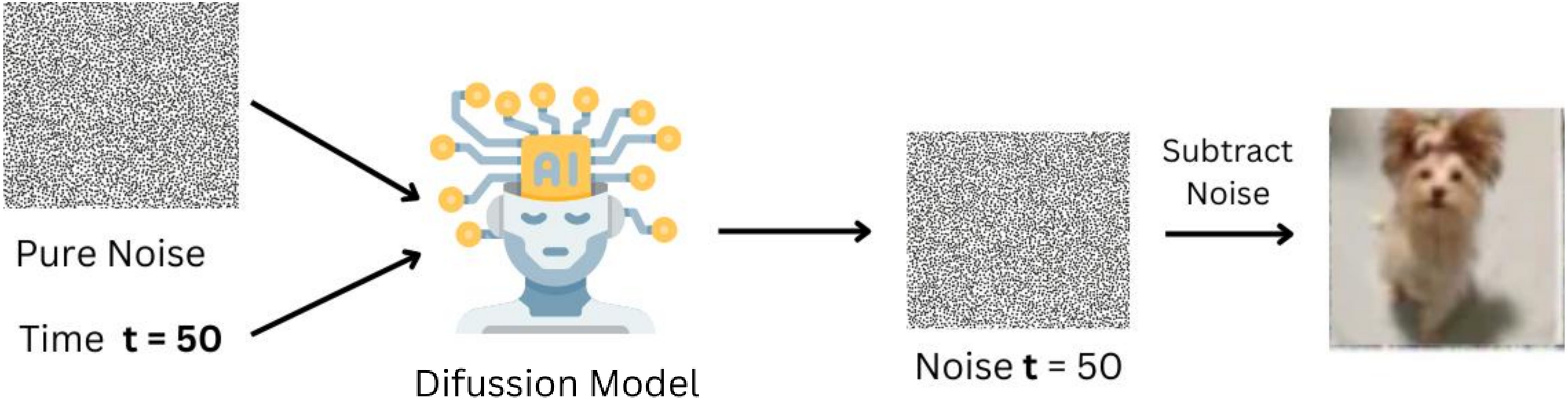
Stable Diffusion - Loss

$$L_{\text{simple}} = E_{t,x_0,\epsilon}\left[||\epsilon - \epsilon_\theta(x_t, t)||^2\right]$$

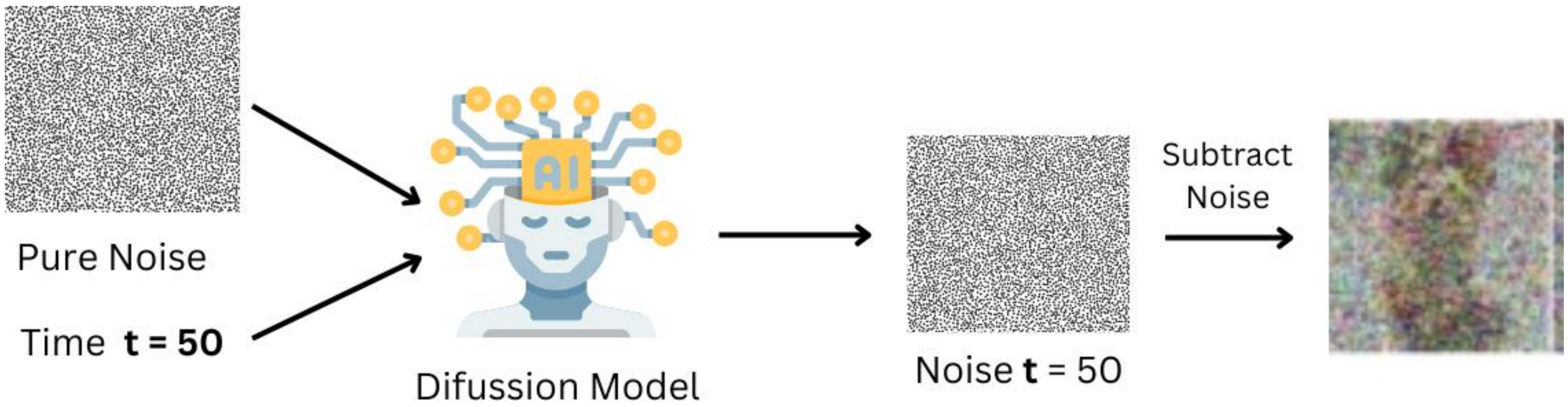
1. The loss from DDPM paper[9]
2. Mean Squared Error between predicted noise and real noise

[9] Ho, Jonathan, Ajay Jain, and Pieter Abbeel. "Denoising diffusion probabilistic models." Advances in neural information processing systems 33 (2020): 6840-6851.

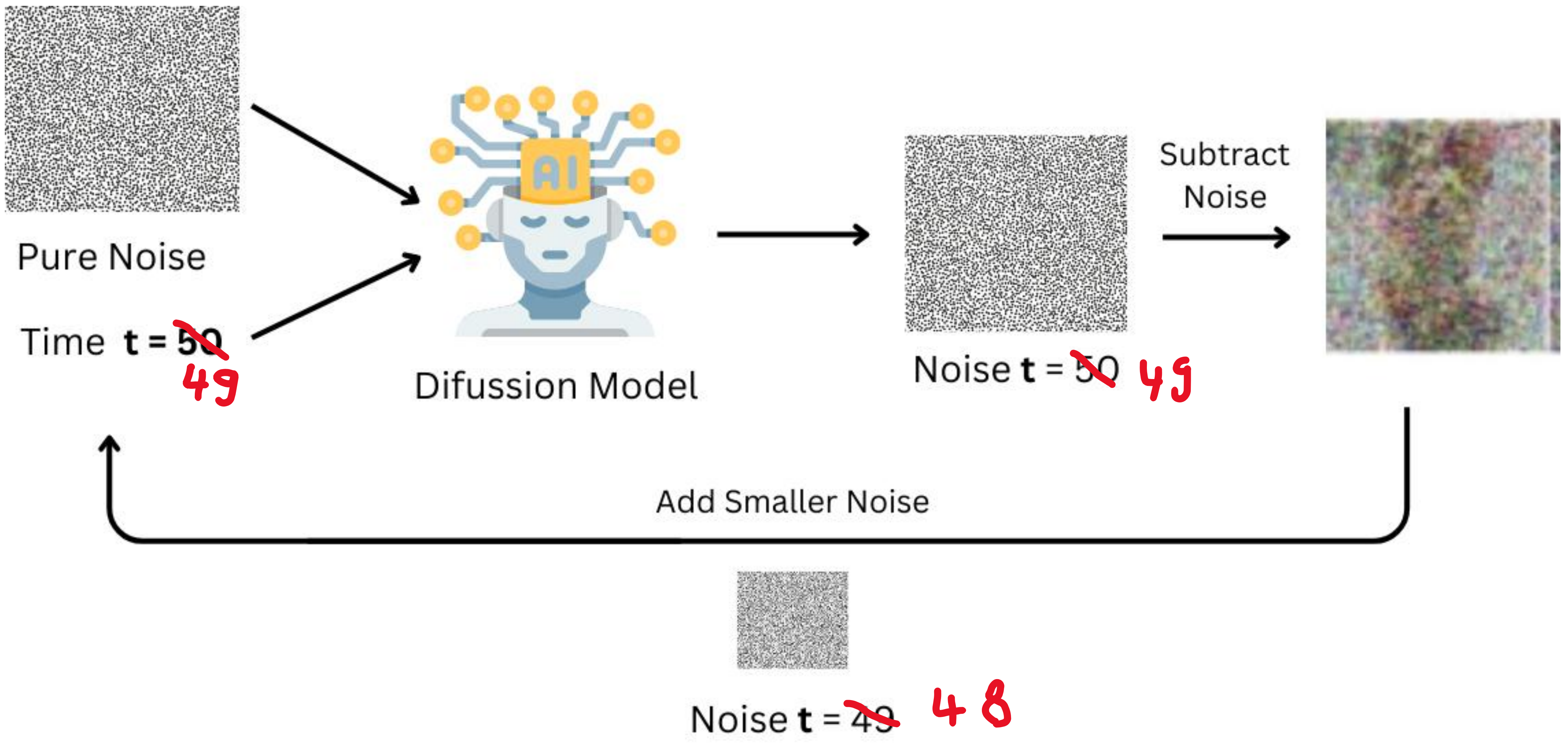
Generation Procedure



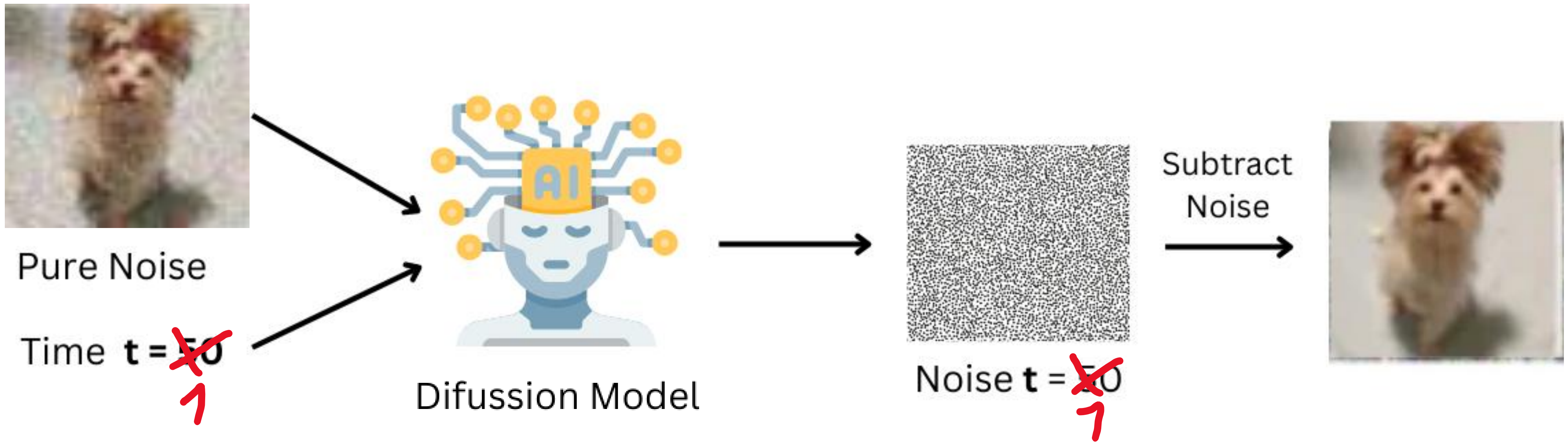
Generation Procedure



Generation Procedure



Generation Procedure



Computerphile Trick

Image from “How I Understand Diffusion Models”- https://www.youtube.com/watch?v=-lz30by8-sU

Image Colorization

- Interesting Git Repo: https://github.com/ErwannMillon/Color-diffusion

Diffusion Timeline

Dec. 2020
Denoising Diffusion Probabilistic Models
UC Berkeley

Feb. 2021
Improved Denoising Diffusion Probabilistic Models
Open AI

Jun. 2021
Diffusion Models Beat GANs on Image Synthesis
Open AI

Dec. 2021
Classifier-Free Diffusion Guidance
Google Research

Mar. 2022
GLIDE: Towards Photorealistic Image Generation and Editing with Text-Guided Diffusion Models
Open AI

Apr. 2022
Hierarchical Text-Conditional Image Generation with CLIP Latents (DALL-E 2)
Open AI

Apr. 2022
High-Resolution Image Synthesis with Latent Diffusion Models
Ludwig Maximilian University
IWR, Heidelberg University
Runway ML

May 2022
Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding (ImageGen)
Google Research

Dec 2022
Stable Diffusion 2.1
Stability AI

Nov. 2023
Adding Conditional Control to Text-to-Image Diffusion Models
Standford University

Improved DDPM[12]

- Learned Noise Schedules
  - Changed fixed variance -> learned (cosine) noise schedule
  - Reduced diffusions steps from 1000 to 50-100
- Classifier Conditioning

[12] Nichol, Alexander Quinn, and Prafulla Dhariwal. "Improved denoising diffusion probabilistic models." International conference on machine learning. PMLR, 2021.

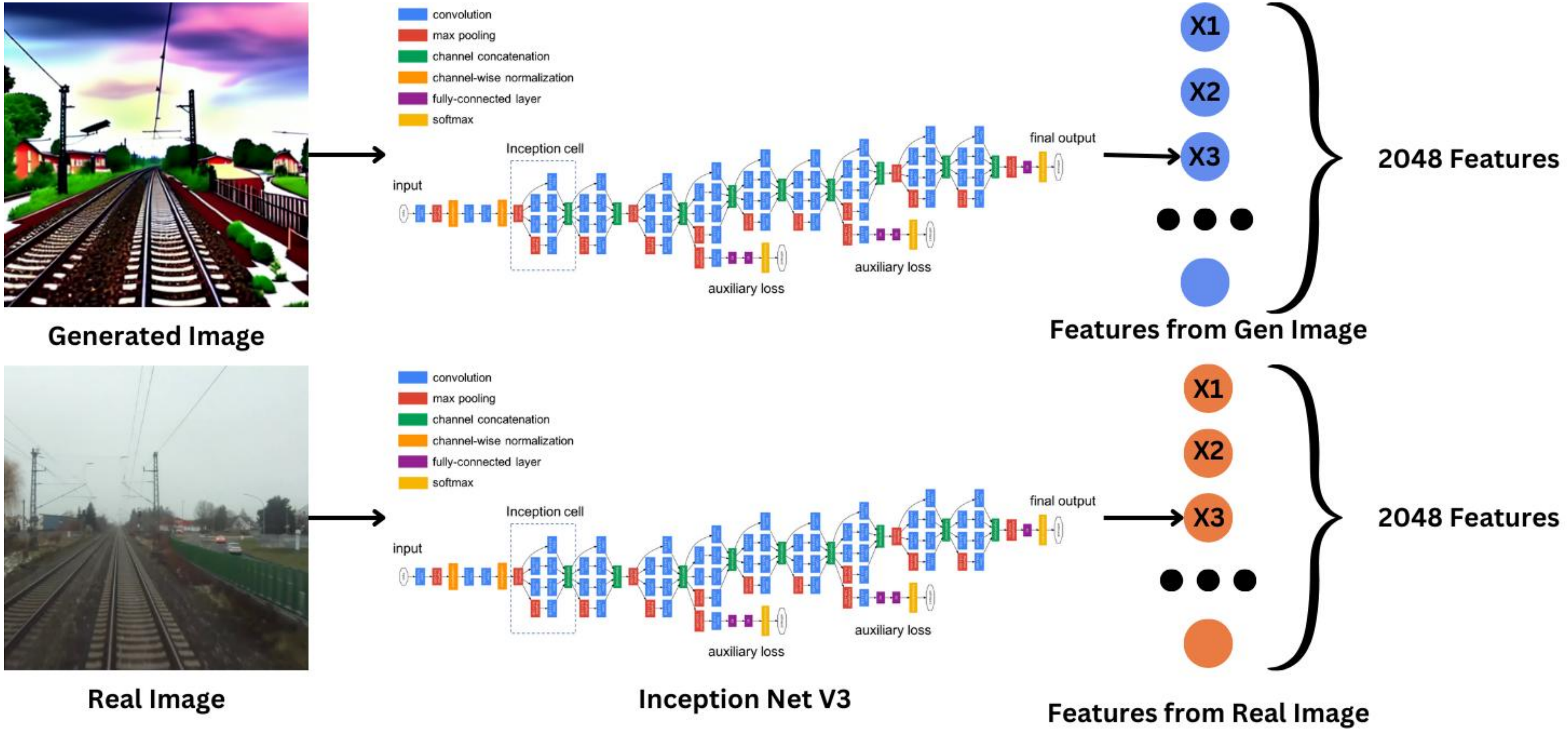
Diffusion Models Beat GANs on Image Synthesis [13]

- Demonstrating Superiority of Diffusion over GANs
- Architecture Improvements. U-Net with self-attention
- Classifier Guidance
  - Required pre-trained classifier

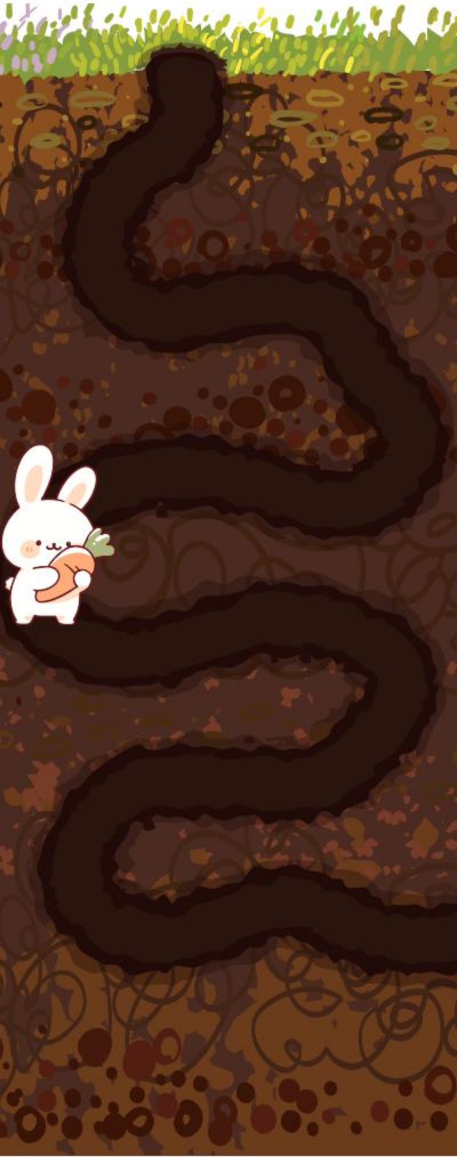
[13] Dhariwal, Prafulla, and Alexander Nichol. "Diffusion models beat gans on image synthesis." Advances in neural information processing systems 34 (2021): 8780-8794.

How do we evaluate Image Generation?

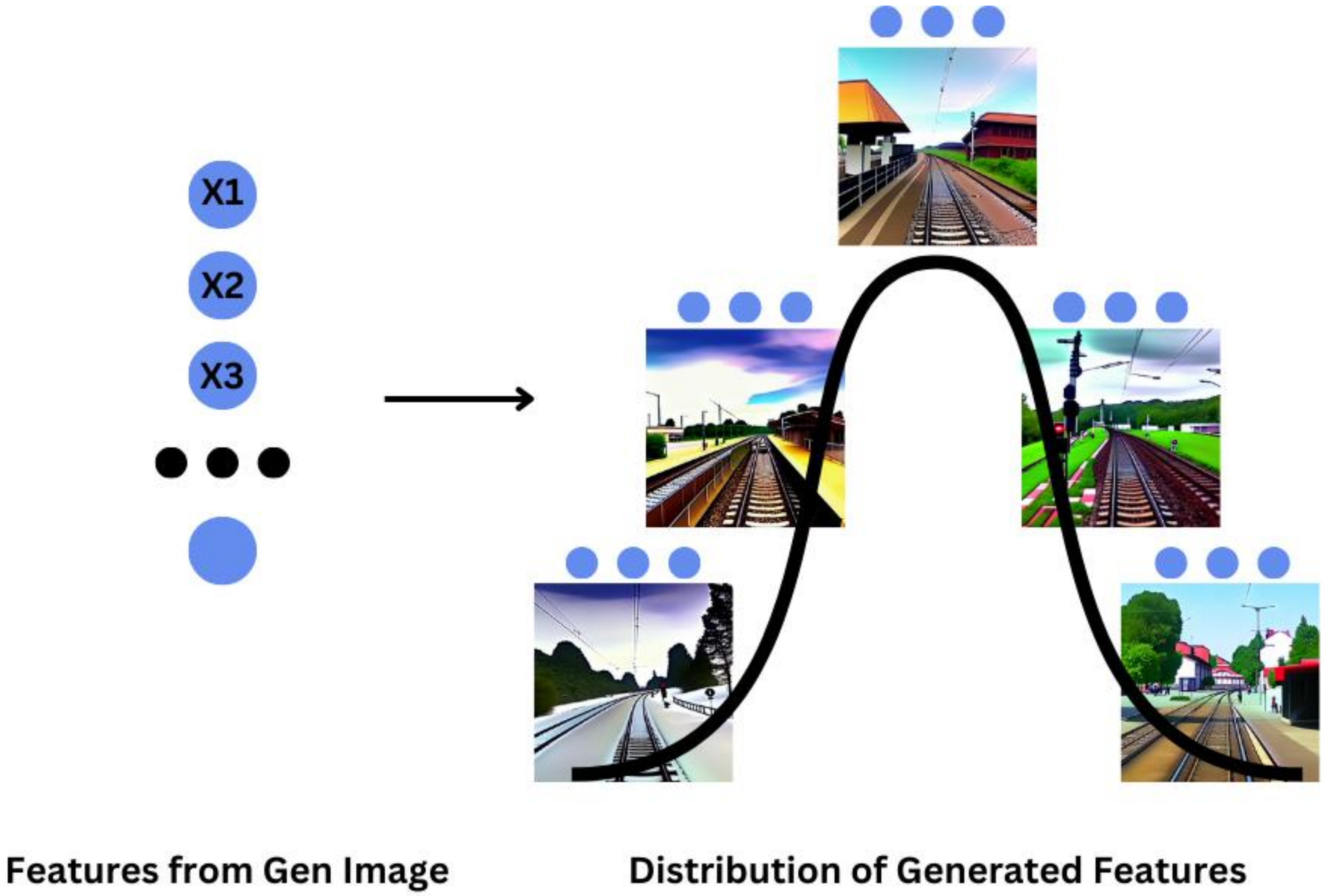
Evaluation - Fréchet Inception Distance



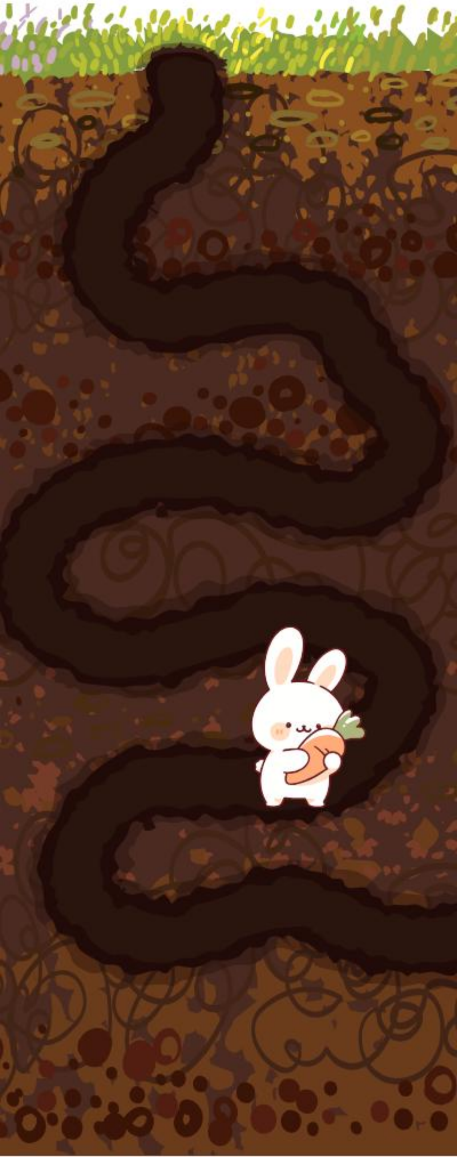
Rabbit Hole
Progress Tracking



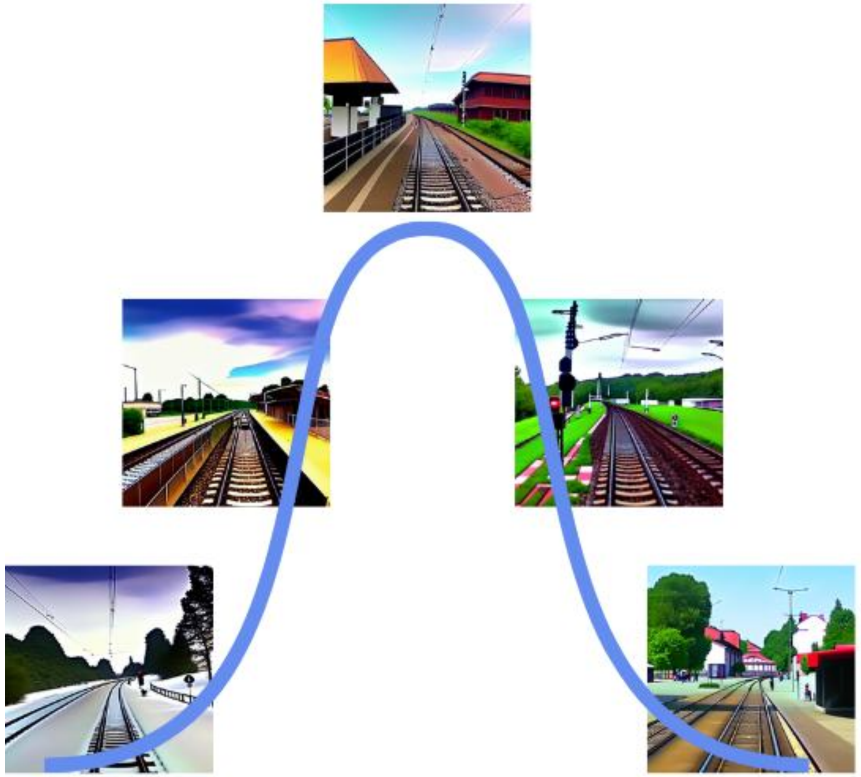
Evaluation - Fréchet Inception Distance



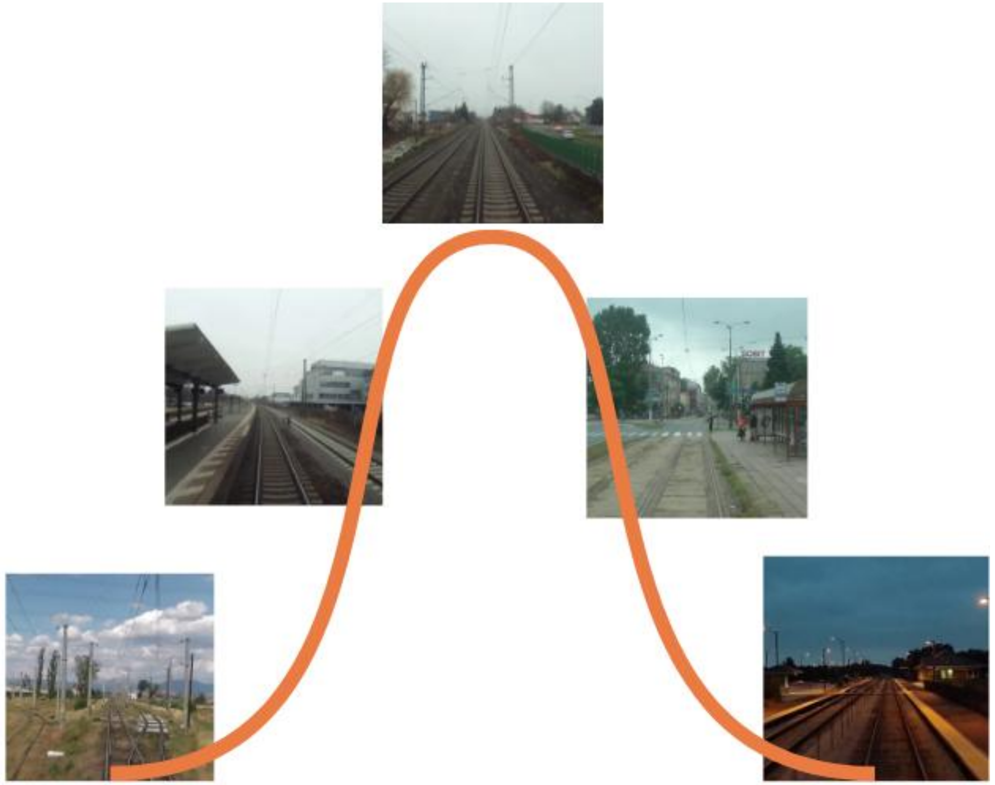
Rabbit Hole
Progress Tracking



Evaluation - Fréchet Inception Distance

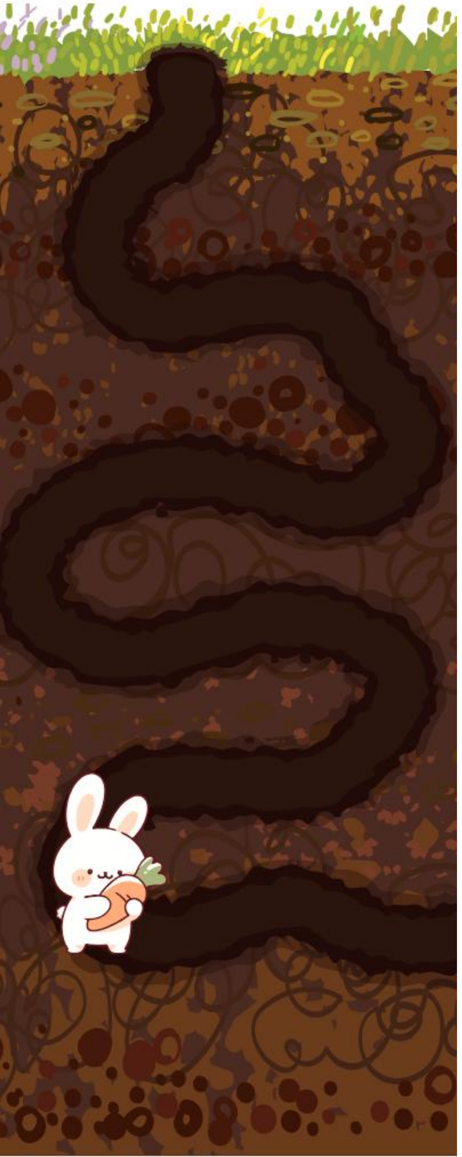


Distribution of Generated Features



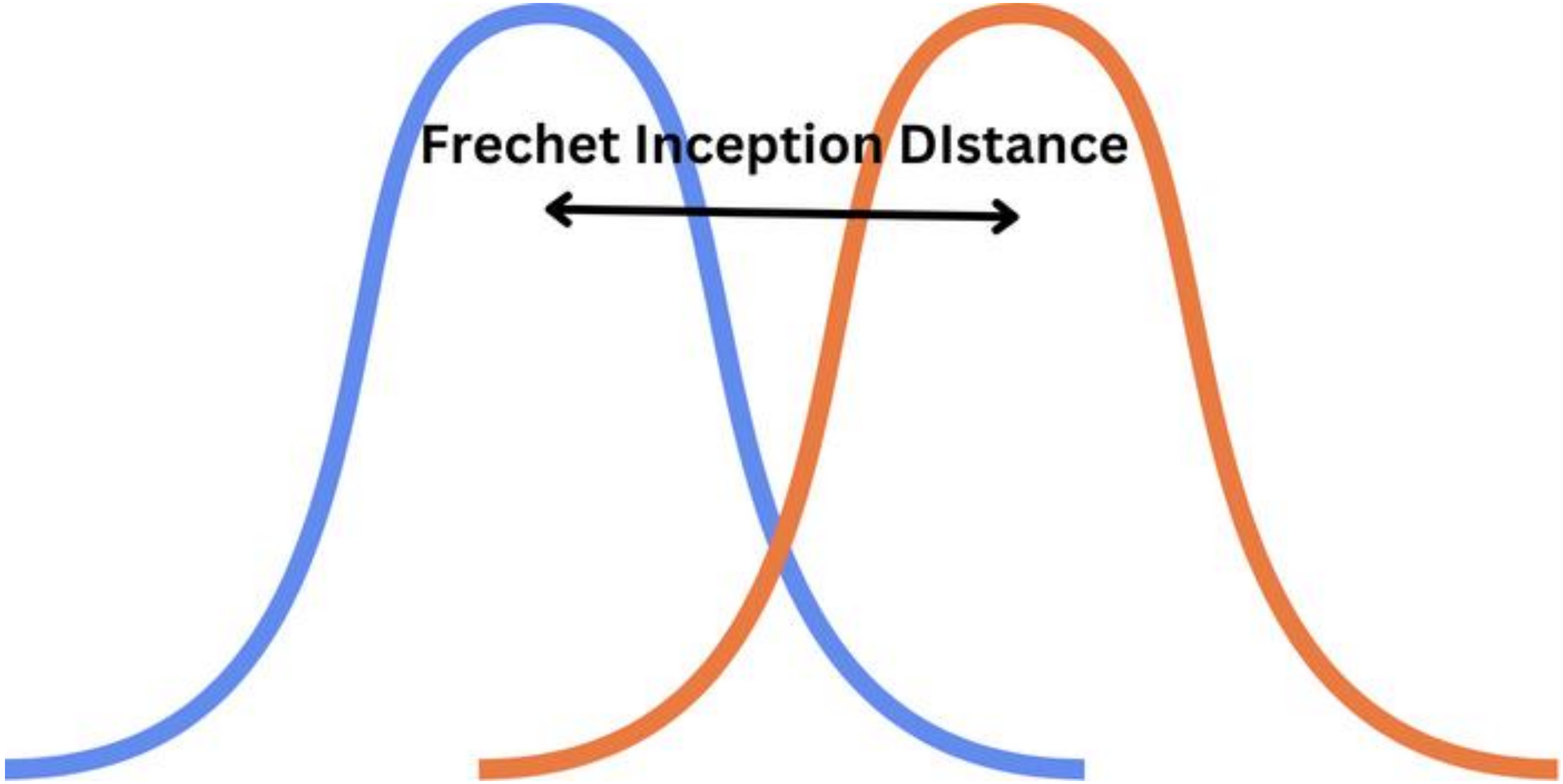
Distribution of Real Features

Rabbit Hole
Progress Tracking



Evaluation - Fréchet Inception Distance

$$FID = \|\mu_r - \mu_g\|^2 + T_r(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$$



Rabbit Hole
Progress Tracking

Evaluation - Fréchet Inception Distance

Instructions

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Diffusion Models Beat GANs on Image Synthesis

Model	FID
LSUN Bedrooms 256×256	
DCTransformer† [42]	6.40
DDPM [25]	4.89
IDDPM [43]	4.24
StyleGAN [27]	2.35
ADM (dropout)	**1.90**
LSUN Horses 256×256	
StyleGAN2 [28]	3.84
ADM	2.95
ADM (dropout)	**2.57**
LSUN Cats 256×256	
DDPM [25]	17.1
StyleGAN2 [28]	7.25
ADM (dropout)	**5.57**
ImageNet 64×64	
BigGAN-deep* [5]	4.06
IDDPM [43]	2.92
ADM	2.61
ADM (dropout)	**2.07**

Model	FID
ImageNet 128×128	
BigGAN-deep [5]	6.02
LOGAN† [68]	3.36
ADM	5.91
ADM-G (25 steps)	5.98
ADM-G	**2.97**
ImageNet 256×256	
DCTransformer† [42]	36.51
VQ-VAE-2†‡ [51]	31.11
IDDPM‡ [43]	12.26
SR3†‡ [53]	11.30
BigGAN-deep [5]	6.95
ADM	10.94
ADM-G (25 steps)	5.44
ADM-G	**4.59**
ImageNet 512×512	
BigGAN-deep [5]	8.43
ADM	23.24
ADM-G (25 steps)	8.41
ADM-G	**7.72**

Diffusion Models Beat GANs on Image Synthesis

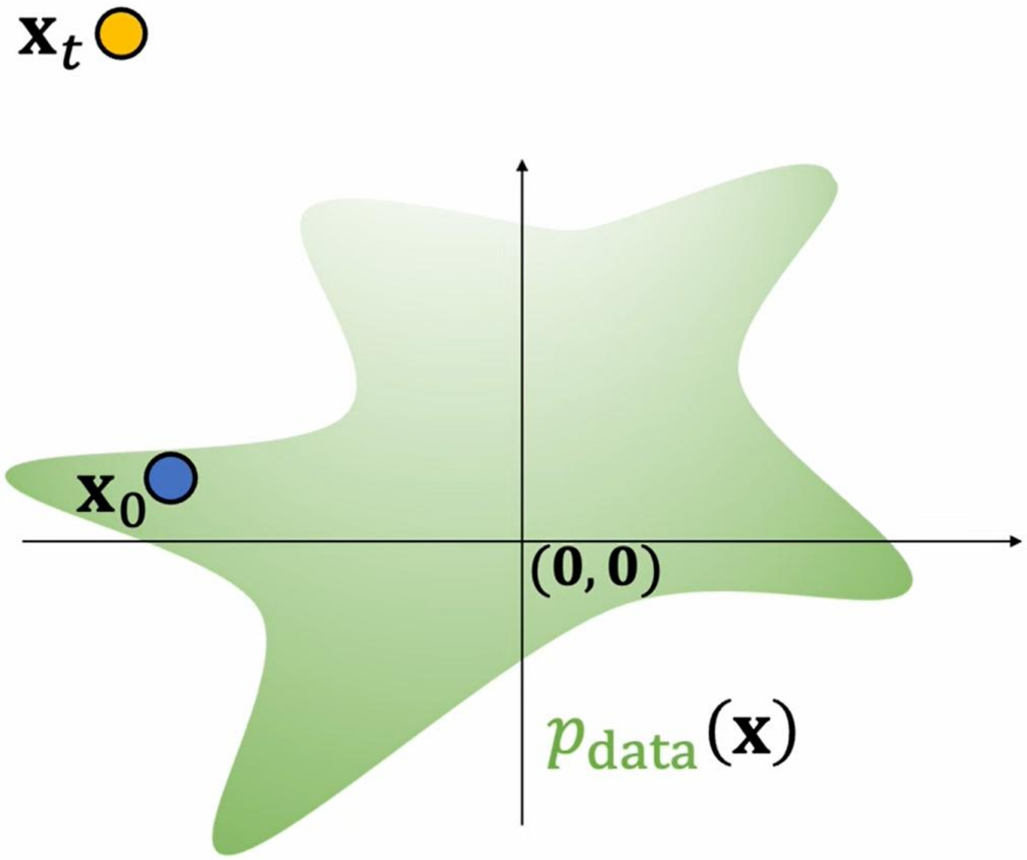


Image from "How I Understand Diffusion Models" - https://www.youtube.com/watch?v=i2qSxMVeVLI

Diffusion Models Beat GANs on Image Synthesis

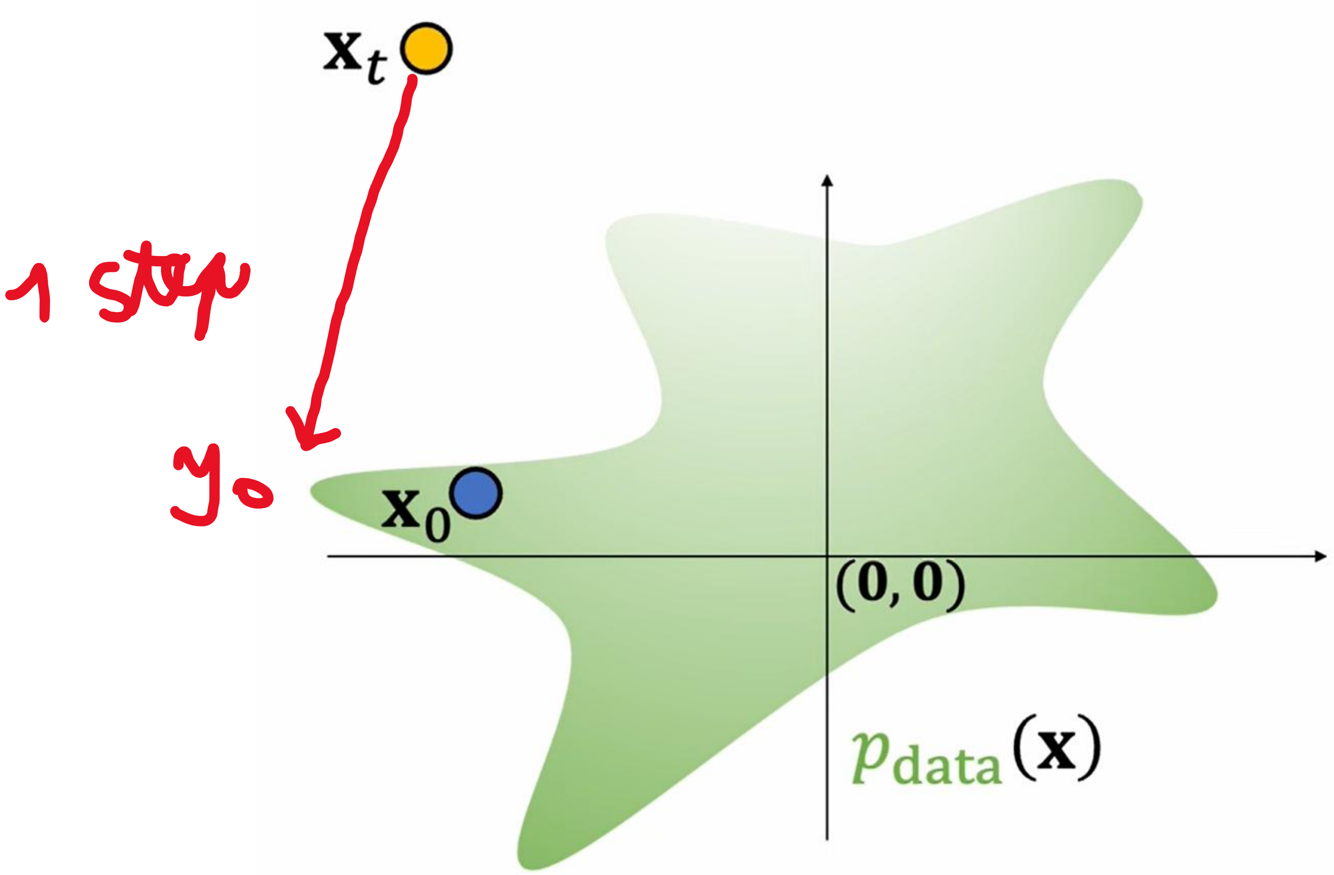


Image from “How I Understand Diffusion Models” - https://www.youtube.com/watch?v=i2qSxMVeVLI

Diffusion Models Beat GANs on Image Synthesis

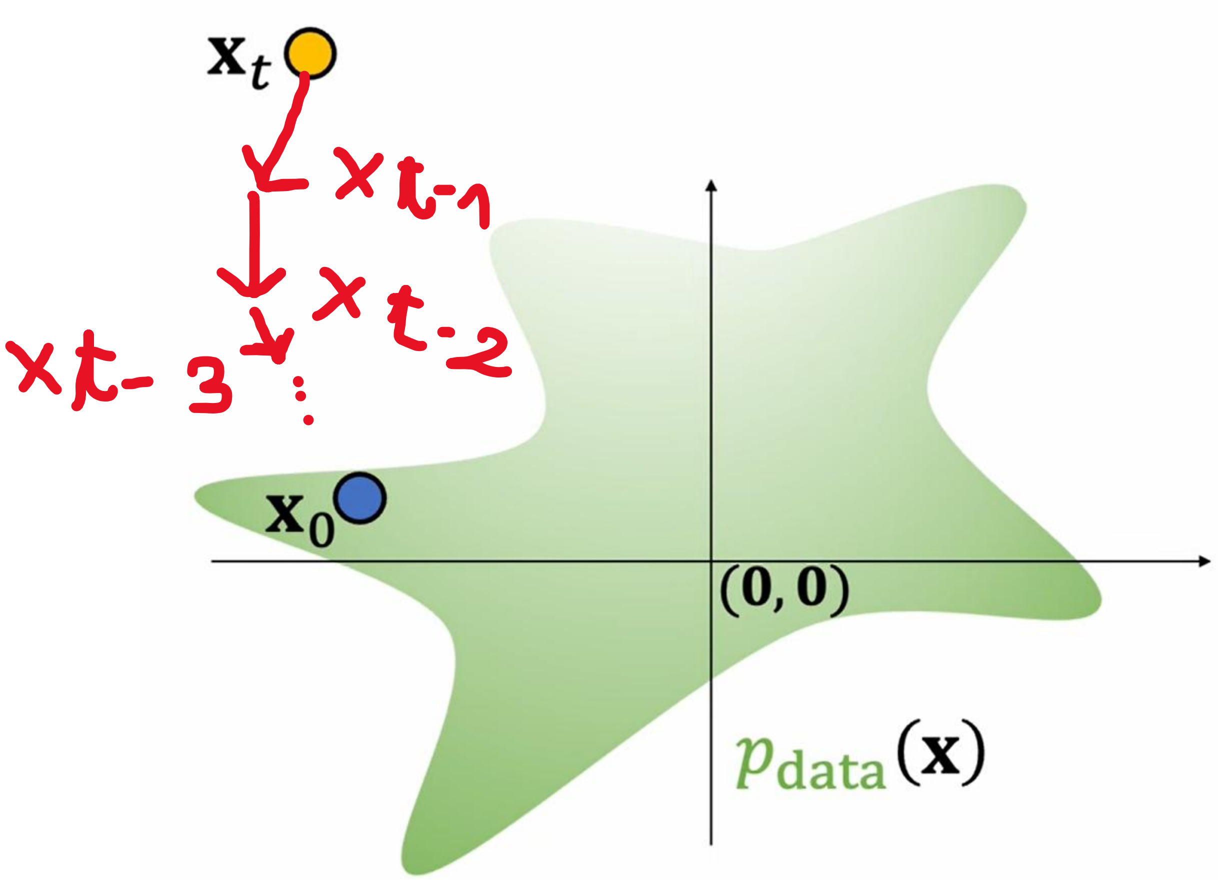


Image from “How I Understand Diffusion Models” - https://www.youtube.com/watch?v=i2qSxMVeVLI

Classifier Guidance

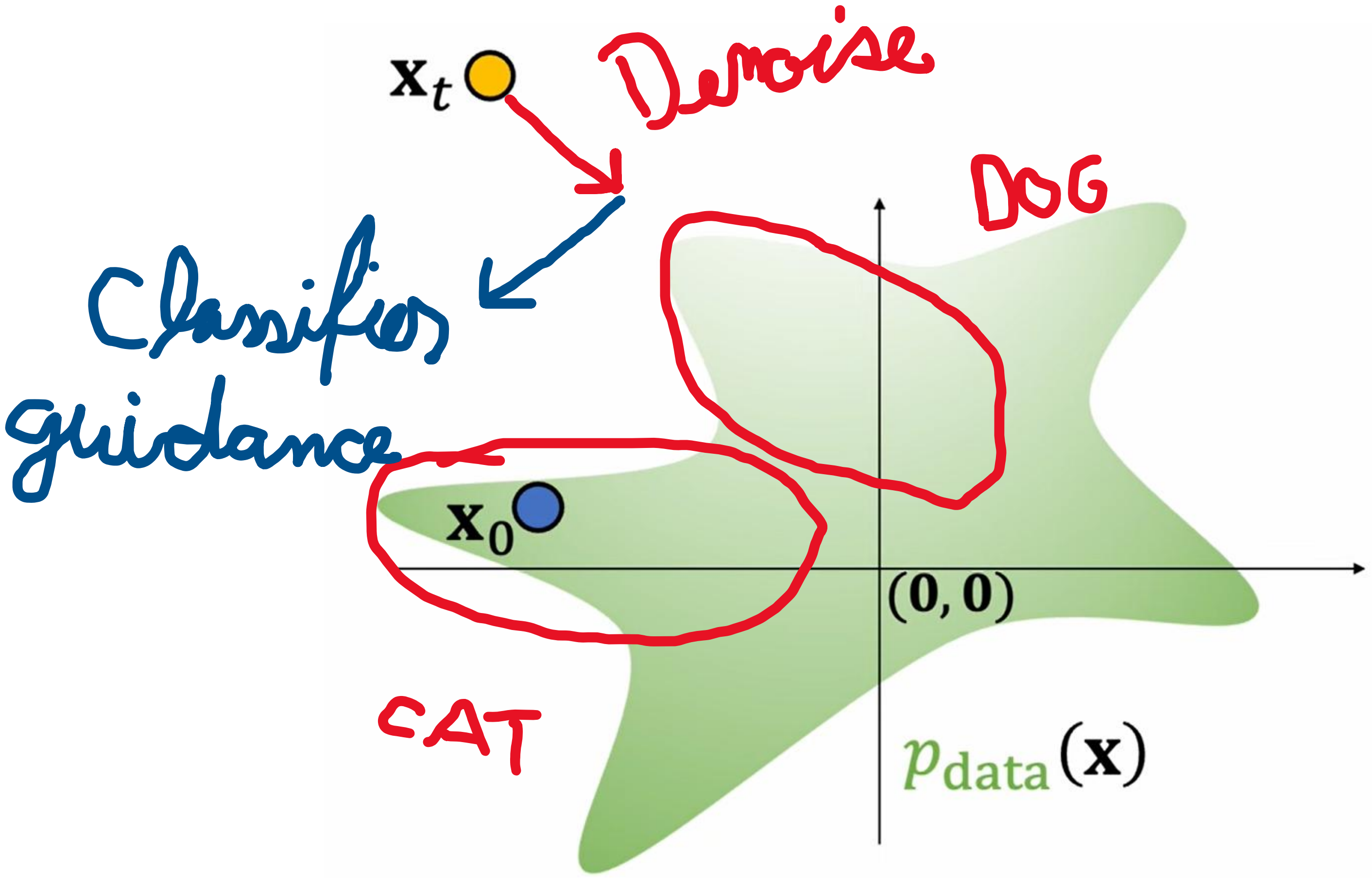


Image from “How I Understand Diffusion Models” - https://www.youtube.com/watch?v=i2qSxMVeVLI

Classifier-Free Diffusion Guidance [14]

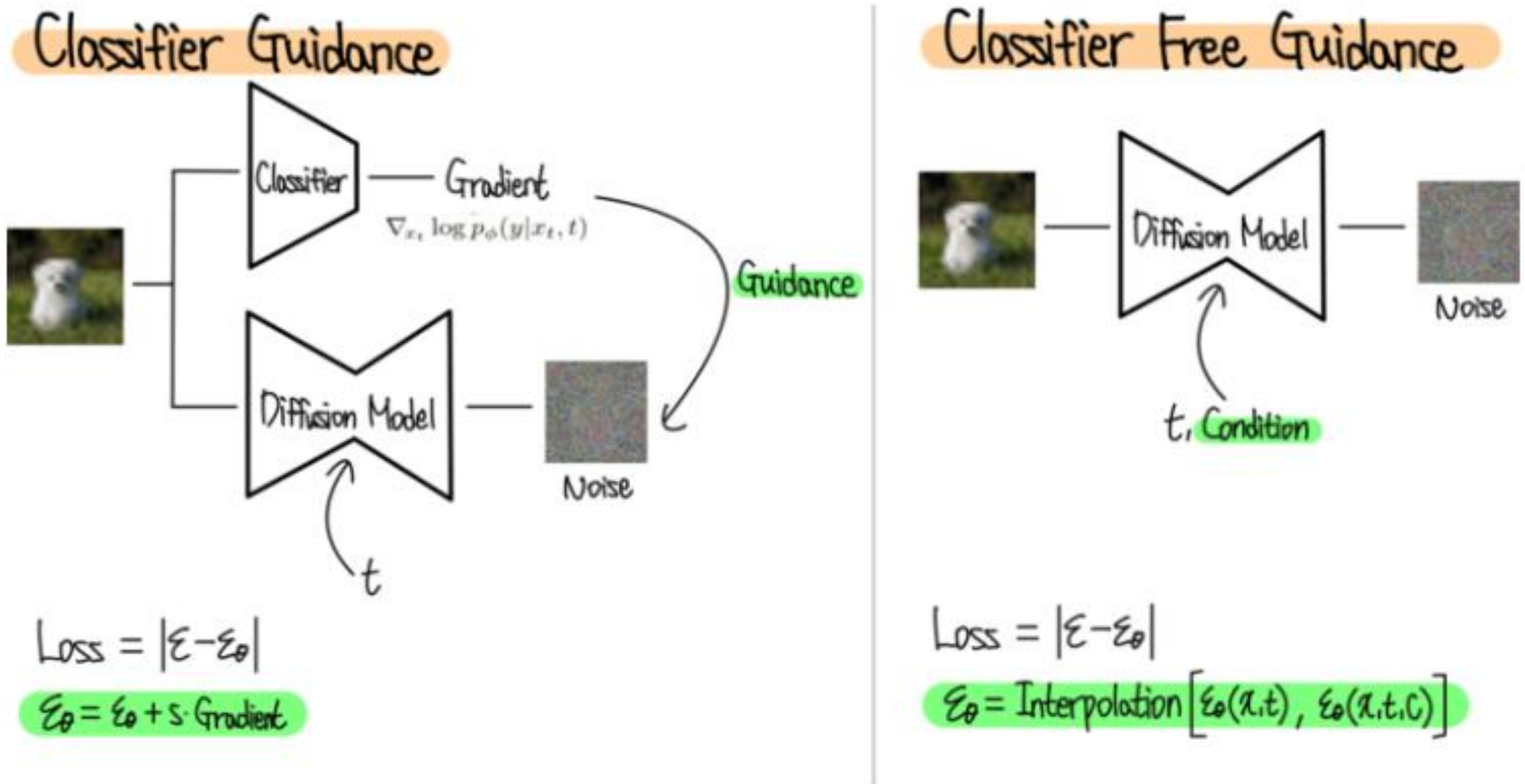


Image from https://ffighting.net/deep-learning-paper-review/diffusion-model/classifier-free-guidance/

[14] Ho, Jonathan, and Tim Salimans. 'Classifier-Free Diffusion Guidance'. NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications, 2021, https://openreview.net/forum?id=qw8AKxfYbI.

GLIDE: Towards Photorealistic Image Generation and Editing with Text-Guided Diffusion Models [15]

- Text-Guided Image Generation
- CLIP [16] guidance
- Classifier guidance yielded better results

[15] Nichol, Alex, et al. "Glide: Towards photorealistic image generation and editing with text-guided diffusion models." arXiv preprint arXiv:2112.10741 (2021).

[16] Radford, Alec, et al. "Learning transferable visual models from natural language supervision." International conference on machine learning. PMLR, 2021.

CLIP

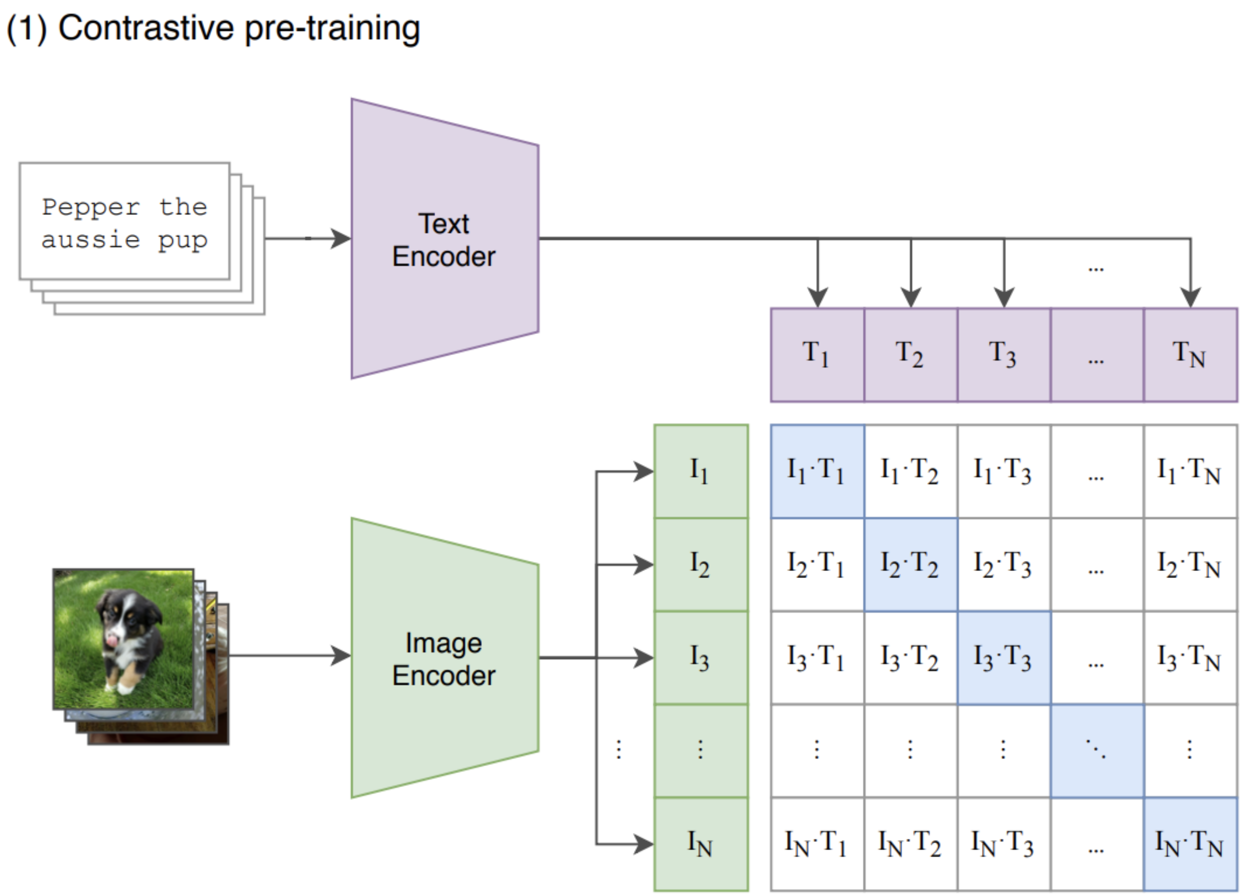


Image from Radford, Alec, et al. "Learning transferable visual models from natural language supervision." International conference on machine learning. PMLR, 2021.

"a hedgehog using a calculator"

"a corgi wearing a red bowtie and a purple party hat"

"robots meditating in a vipassana retreat"

"a fall landscape with a small cottage next to a lake"

"a surrealist dream-like oil painting by salvador dalí of a cat playing checkers"

"a professional photo of a sunset behind the grand canyon"

"a high-quality oil painting of a psychedelic hamster dragon"

"an illustration of albert einstein wearing a superhero costume"

"a boat in the canals of venice"

"a painting of a fox in the style of starry night"

"a red cube on top of a blue cube"

"a stained glass window of a panda eating bamboo"

"a crayon drawing of a space elevator"

"a futuristic city in synthwave style"

"a pixel art corgi pizza"

"a fog rolling into new york"

Hierarchical Text-Conditional Image Generation with CLIP Latents (UnCLIP \ DALL-E 2) [17]

- Text embeddings of CLIP (as opposed to transfer text embeddings)
- Feed image encodings to the decoder

[17] Ramesh, Aditya, et al. "Hierarchical text-conditional image generation with clip latents." arXiv preprint arXiv:2204.06125 1.2 (2022): 3.

UnCLIP \ DALL-E 2

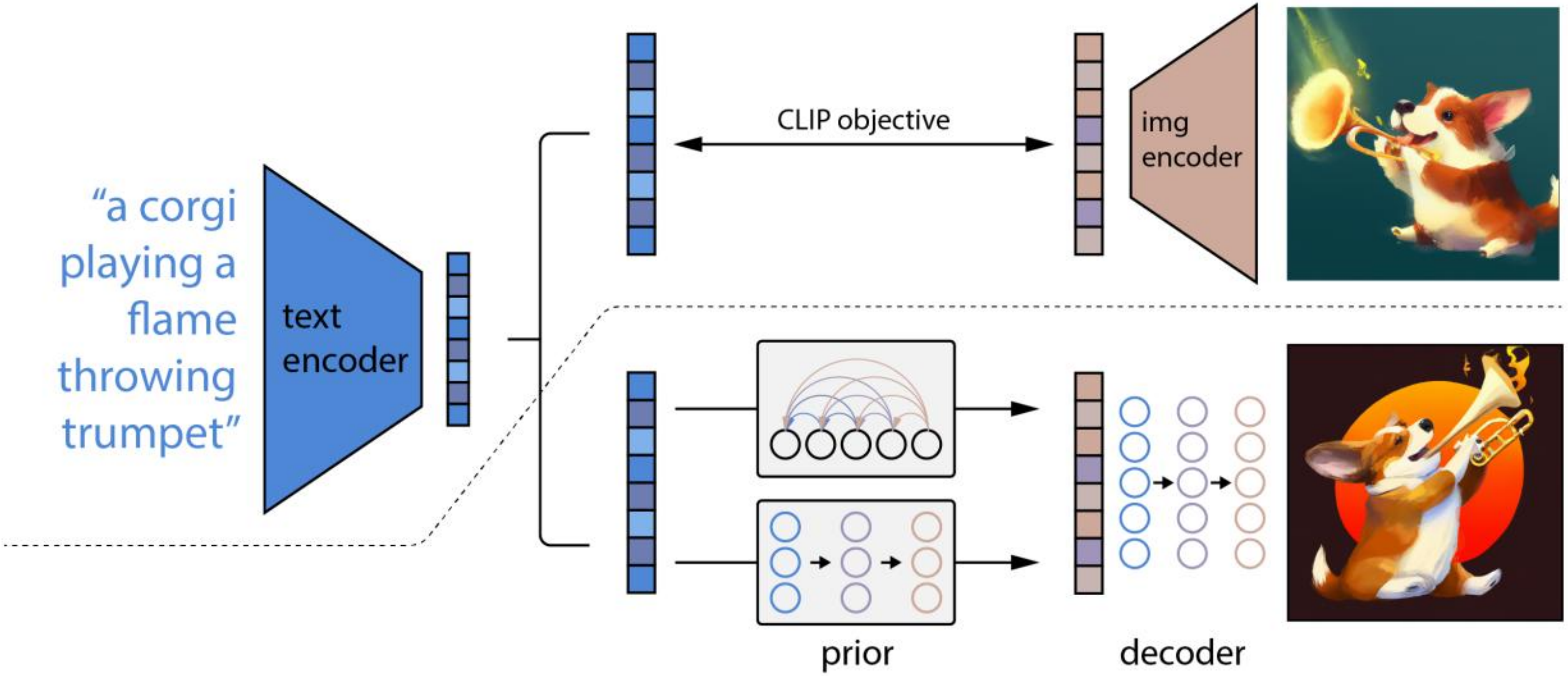


Image from Ramesh, Aditya, et al. "Hierarchical text-conditional image generation with clip latents." arXiv preprint arXiv:2204.06125 1.2 (2022): 3..


vibrant portrait painting of Salvador Dalí with a robotic half face


a shiba inu wearing a beret and black turtleneck


a close up of a handpalm with leaves growing from it


an espresso machine that makes coffee from human souls, artstation

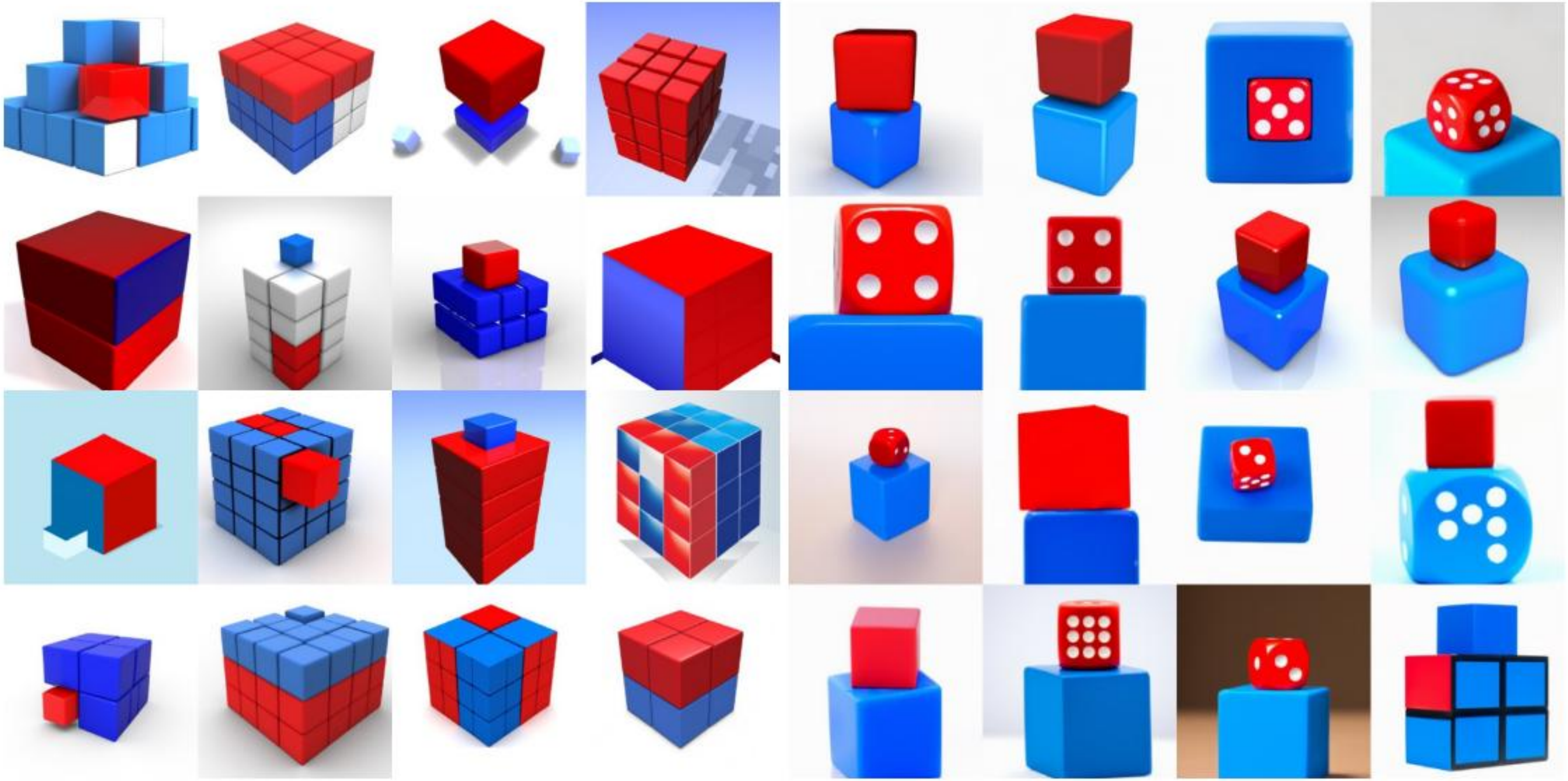

panda mad scientist mixing sparkling chemicals, artstation


a corgi's head depicted as an explosion of a nebula


a dolphin in an astronaut suit on saturn, artstation


a propaganda poster depicting a cat dressed as french emperor napoleon holding a piece of cheese


a teddy bear on a skateboard in times square

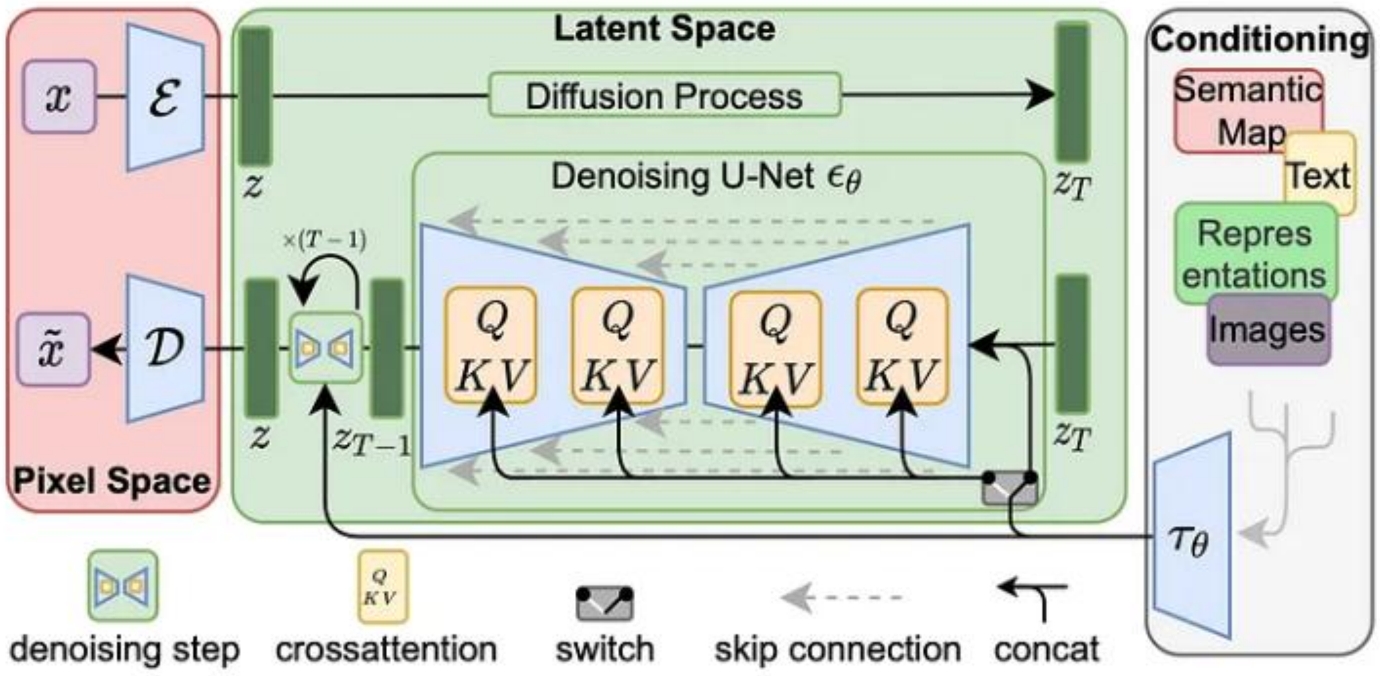


(a) unCLIP (b) GLIDE

Image from Ramesh, Aditya, et al. "Hierarchical text-conditional image generation with clip latents." arXiv preprint arXiv:2204.06125 1.2 (2022): 3..

High-Resolution Image Synthesis with Latent Diffusion Models [18]

- Diffusion in Latent Space Instead of Pixel Space
- LDM allow high-resolution image synthesis (1024x1024)



[18] Rombach, Robin, et al. "High-resolution image synthesis with latent diffusion models." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2022.

Input	**ours** ($f = 4$) PSNR: 27.4 R-FID: 0.58	**DALL-E** ($f = 8$) PSNR: 22.8 R-FID: 32.01	**VQGAN** ($f = 16$) PSNR: 19.9 R-FID: 4.98

Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding [19]

- Google ImageGen: https://imagen.research.google/
- Enhanced Photorealism
- Advanced Language Understanding (PaLM)
- Multiscale Diffusion

[19] Saharia, Chitwan, et al. "Photorealistic text-to-image diffusion models with deep language understanding." Advances in neural information processing systems 35 (2022): 36479-36494.

Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding [19]

Model	FID-30K	Zero-shot FID-30K
AttnGAN [76]	35.49	
DM-GAN [83]	32.64	
DF-GAN [69]	21.42	
DM-GAN + CL [78]	20.79	
XMC-GAN [81]	9.33	
LAFITE [82]	8.12	
Make-A-Scene [22]	7.55	
DALL-E [53]		17.89
LAFITE [82]		26.94
GLIDE [41]		12.24
DALL-E 2 [54]		10.39
Imagen (Our Work)		**7.27**

Sprouts in the shape of text 'Imagen' coming out of a fairytale book.

A photo of a Shiba Inu dog with a backpack riding a bike. It is wearing sunglasses and a beach hat.

A high contrast portrait of a very happy fuzzy panda dressed as a chef in a high end kitchen making dough. There is a painting of flowers on the wall behind him.

Teddy bears swimming at the Olympics 400m Butterfly event.

A cute corgi lives in a house made out of sushi.

A cute sloth holding a small treasure chest. A bright golden glow is coming from the chest.

A brain riding a rocketship heading towards the moon.

A dragon fruit wearing karate belt in the snow.

A strawberry mug filled with white sesame seeds. The mug is floating in a dark chocolate sea.

Instructions

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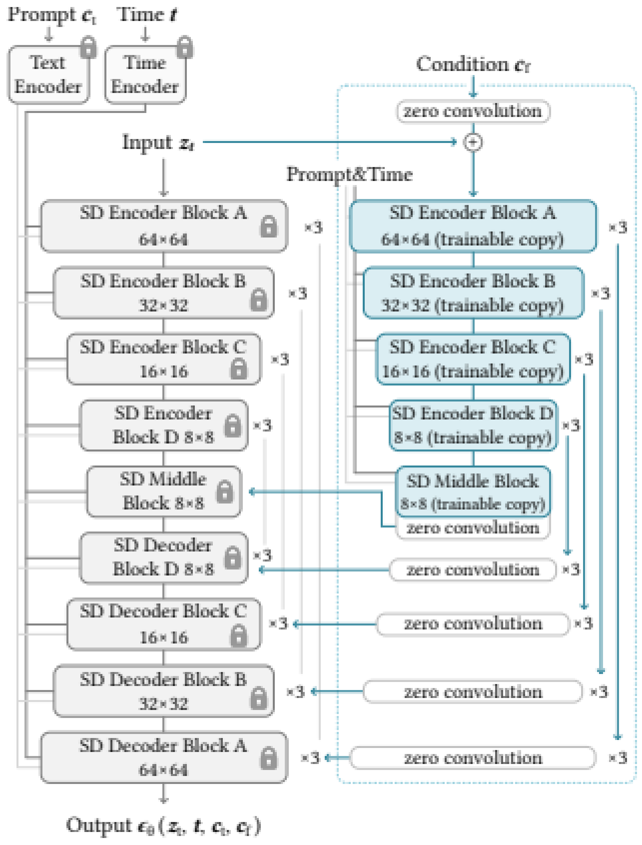
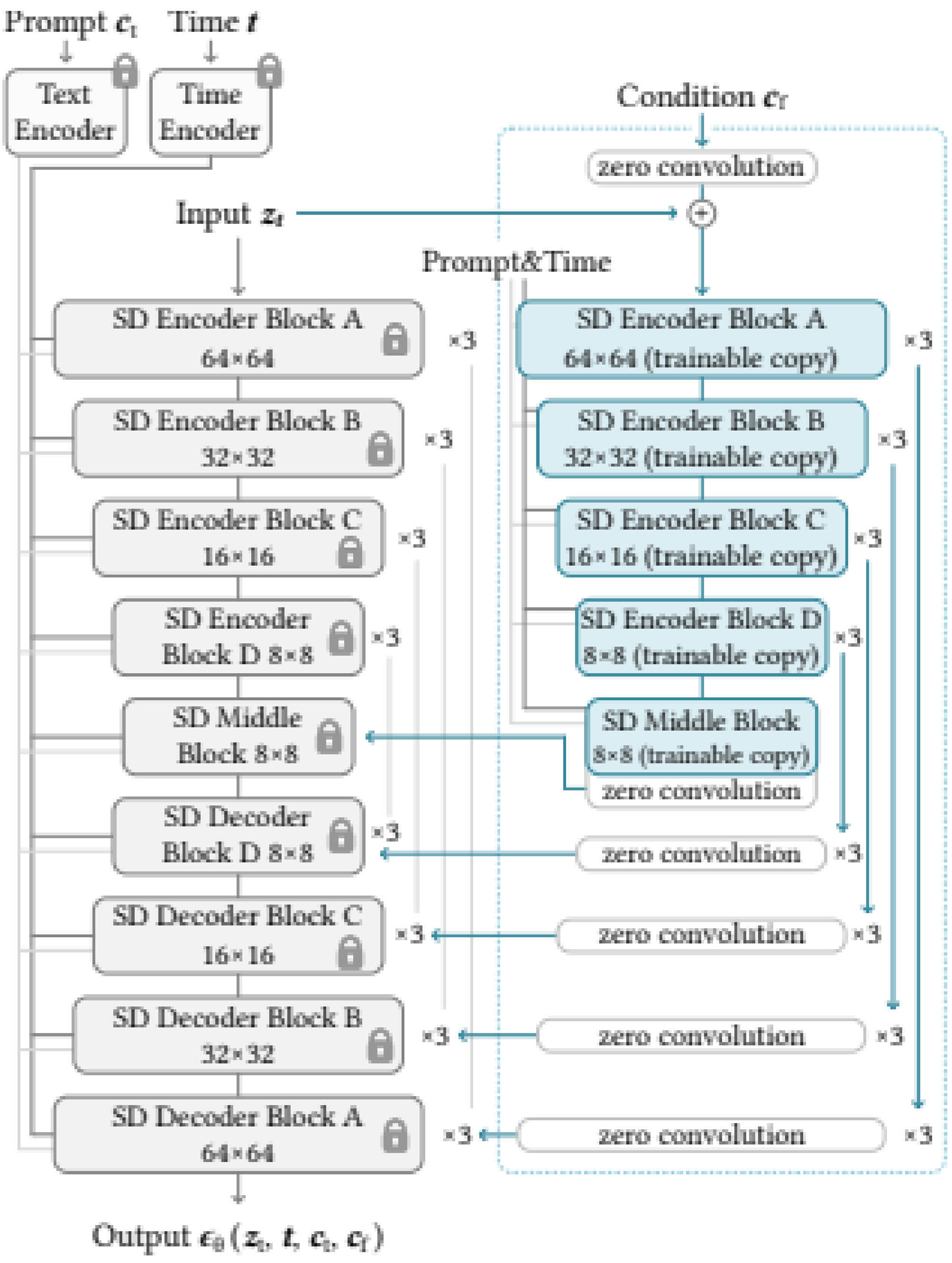
Adding Conditional Control to Text-to-Image Diffusion Models [20]



(a) Stable Diffusion (b) ControlNet

[20] Zhang, Lvmin, Anyi Rao, and Maneesh Agrawala. "Adding conditional control to text-to-image diffusion models." Proceedings of the IEEE/CVF International Conference on Computer Vision. 2023.



(a) Stable Diffusion (b) ControlNet


Input Canny edge
Default
“masterpiece of fairy tale, giant deer, golden antlers”
“..., quaint city Galic”
Input human pose
Default
“chef in kitchen”
“Lincoln statue”

Other recent advancements

- DALL-E 3 (late 2023) – Multi- Modal
- Diffusion Transformer Models
  - SORA – Video Generation: https://openai.com/index/sora/
  - Stable Diffusion 3 – Stability AI
- ImageGen 3
- Nano Banana
- Control Net Variants
- DeepCache: Accelerating Diffusion Models for Free [21]

[21] Ma, Xinyin, Gongfan Fang, and Xinchao Wang. "Deepcache: Accelerating diffusion models for free." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2024.

Video Generation

Instructions

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Enter the code

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Or use QR code

Final Question

Instructions

Go to

www.menti.com

Enter the code

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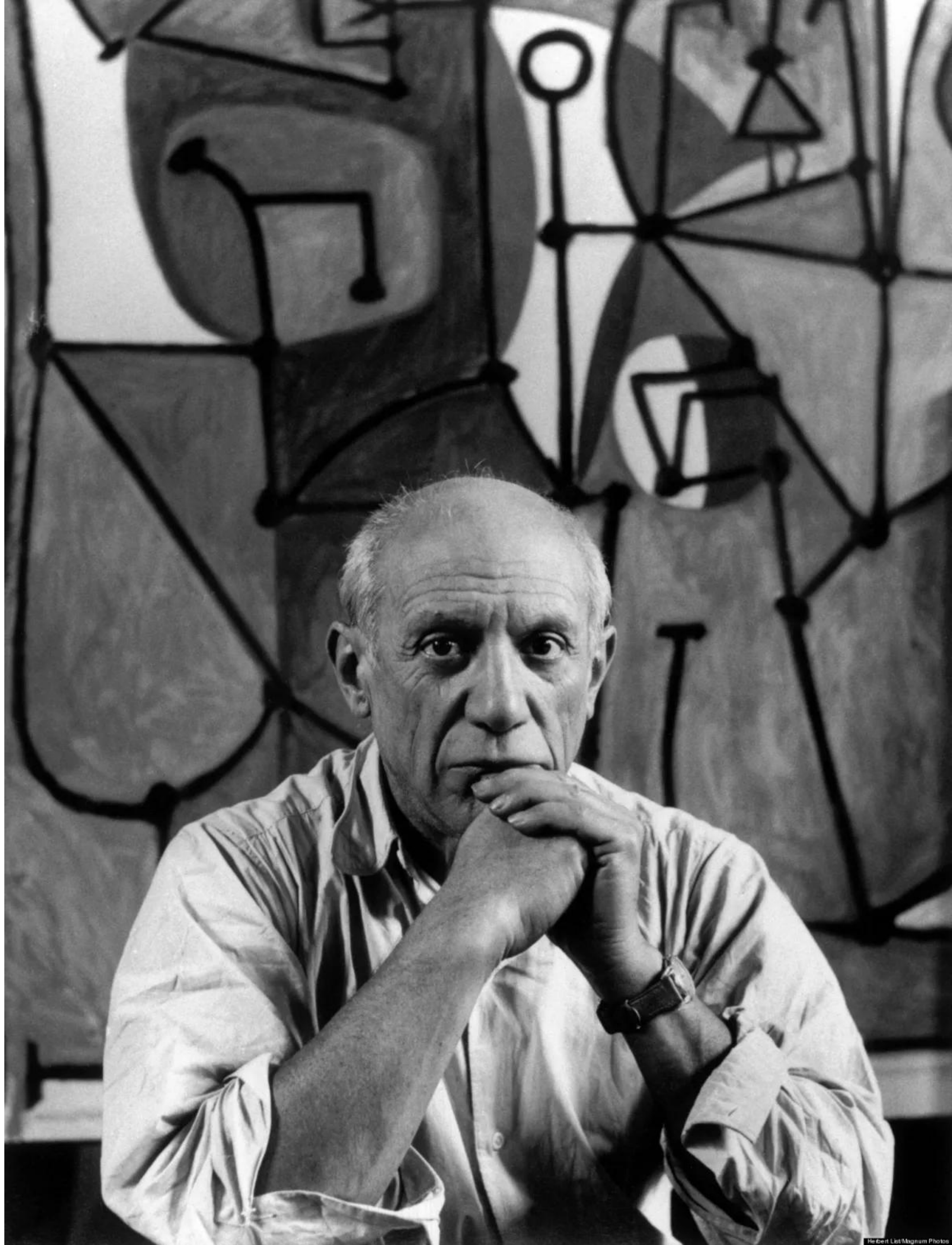
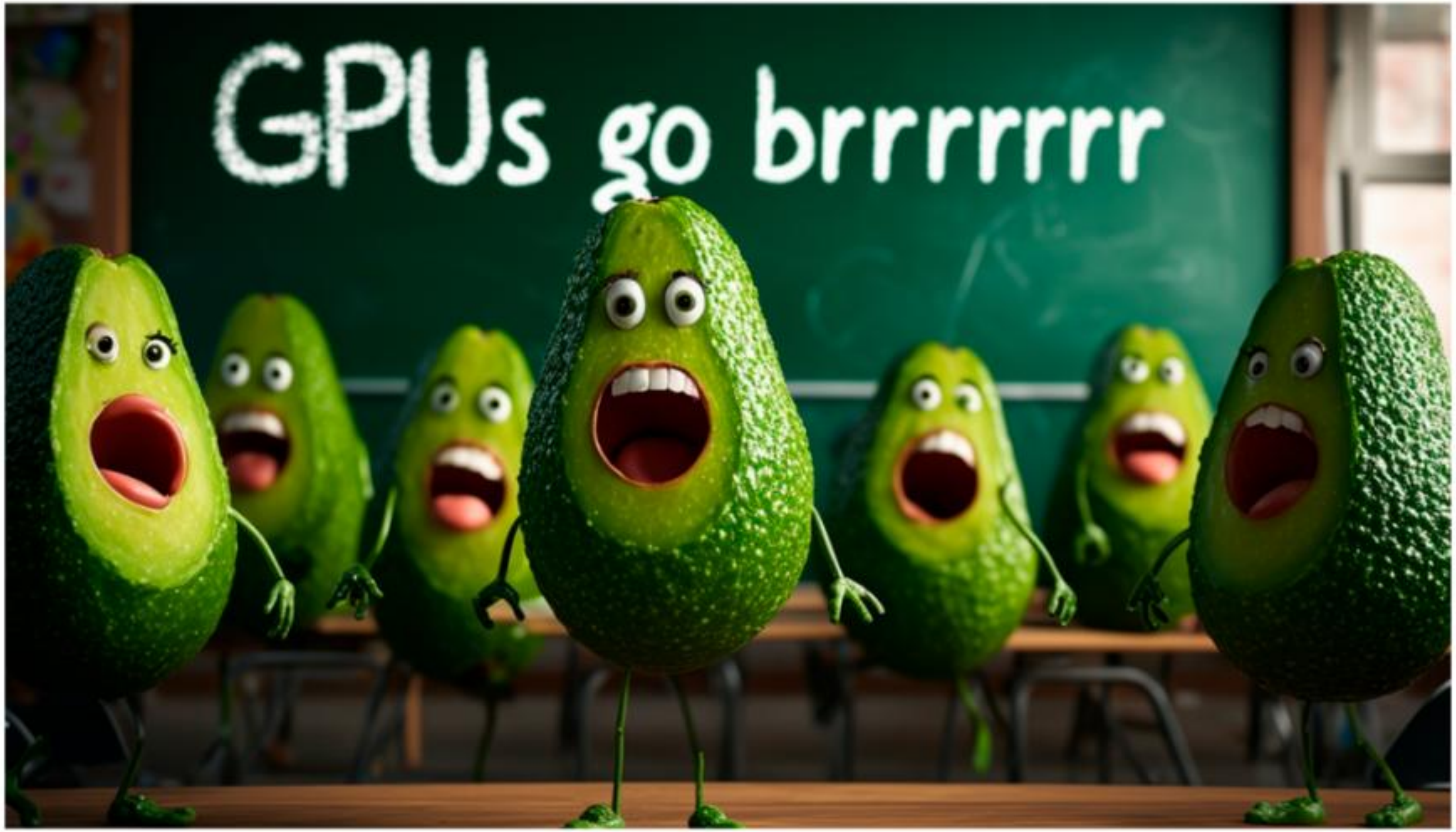
Or use QR code

"Good artists copy,
great artists steal."
- Pablo Picasso

Question: Are Stable Diffusion models good or great artists? Do they copy or do they steal?





Prompt: A surreal and humorous scene in a classroom with the words 'GPUs go brrrrrr' written in white chalk on a blackboard. In front of the blackboard, a group of students are celebrating. These students are uniquely depicted as avocados, complete with little arms and legs, and faces showing expressions of joy and excitement. The scene captures a playful and imaginative atmosphere, blending the concept of a traditional classroom with the whimsical portrayal of avocado students.

Image from Stablity AI – Stable Diffusion 3 https://stability.ai/news/stable-diffusion-3-research-paper